

PROFESSIONAL ANALOG LIVE SOUND ENGINEERING

QAJEELCHA BARSIIISAA FI BARATAA

BOQONNAA 1 — SEENSA SOUND ENGINEERING

1. IBSA BAL'AA

1.1. SOUND ENGINEERING MAAL Jechuudha?

Sound Engineering jechuun sagalee uumamu ykn microphone irraa qabame sirnaan qabachuu, qindeessuu, sadarkaa isaa to'achuu, qulqulleessuu, walitti makuu fi gara dhaggeeffataa geessuu irratti beekumsa fi ogummaa fayyadamuudha.

Sound Engineer immoo nama beekumsa Sound Engineering fi meeshaalee sagalee fayyadamuun sagalee qulqulluu, madaalawaa fi dhaggeeffatamuuf mijataa ta'e qopheessuudha.

Sound Engineer tokko mixer qofa beekuun hin ga'u. Inni microphone, cable, mixer, EQ, amplifier, speaker, monitor, effects, feedback fi karaa guutuu signal ittiin darbu sirriitti hubachuu qaba.

1.2. HOJII SOUND ENGINEER

Sound Engineer yeroo hojii irra jiru:

- Microphone filata.
- Microphone bakka sirrii kaa'a.
- Cable fi connection sakatta'a.
- Mixer qindeessa.
- Gain sirreessa.
- EQ fayyadama.
- AUX fi monitor qindeessa.
- Effects akka Reverb fi Delay to'ata.
- Main Speaker qindeessa.
- Feedback hir'isa.
- Noise adda baasa.
- Soundcheck gaggeessa.
- Yeroo live ta'etti sagalee to'ata.
- Rakkoo sagalee uumamu saffisaan fura.

Kaayyoon hojii isaa sagalee qofa guddisuu miti.

Kaayyoon isaa:

SAGALEE QULQULLUU

+

SAGALEE MADAAALAWAA

+

SAGALEE DHAGGEEFFATAMUUF MIJATAA

1.3. MEESHALEE BU'UURAA SOUND ENGINEERING

Sound Engineer tokko meeshaalee armaan gadii sirriitti beekuu qaba:

1. Microphone
2. Analog Mixer
3. Equalizer (EQ)
4. Compressor
5. Effects Processor
6. Amplifier
7. Main Speaker
8. Monitor Speaker
9. DI Box
10. XLR Cable
11. TRS/TS Cable
12. Speaker Cable
13. Power Distribution
14. Stage Monitor
15. Headphone

Meeshaaleen kun hundi hojii Live Sound keessatti gahee adda addaa qabu.

1.4. SIGNAL FLOW

Signal Flow jechuun karaa signal audio tokko meeshaa tokko irraa ka'ee gara output isaatti ittiin darbuudha.

Fakkeenya:

MICROPHONE
↓
XLR CABLE
↓
MIXER INPUT
↓
GAIN / PREAMP
↓
EQ
↓
FADER
↓
MAIN MIX
↓
AMPLIFIER
↓
SPEAKER

Sound Engineer tokko Signal Flow sirriitti yoo beeke, rakkoo sagalee saffisaan adda baasuu danda'a.

Fakkeenyaaf speaker irraa sagaleen yoo hin dhageenye, battalum speaker balleessaa dha jechuu hin qabu.

Tartiibaan sakatta'uu qaba:

Microphone sagalee nama sirbu qabata.

Microphone sagalee sana gara electrical audio signalitti jijjiira.

XLR cable signal sana gara mixeritti geessa.

Mixer keessatti Sound Engineer sagalee sana qindeessa.

Ergasii signal gara amplifieritti darba.

Amplifier signal sana humneessa.

Speaker immoo signal sana gara sagalee dhagahamuutti jijjiira.

2.2. FAKKEENYA — KEYBOARD

Keyboard

↓

Mixer Input

↓

Channel

↓

Gain

↓

EQ

↓

Fader

↓

Main Mix

↓

Amplifier

↓

Speaker

Keyboard irraa sagaleen dhufu mixer keessatti channel mataa isaa qabaachuu danda'a.

Sound Engineer sagalee keyboard sana vocal, guitar fi meeshaalee biroo waliin madaala.

2.3. FAKKEENYA — GUITAR

Guitar

↓

Pickup / DI Box

↓

Mixer

↓
Channel
↓
EQ
↓
Fader
↓
Main Mix
↓
Speaker

Guitar tokko sirnaan qindeessuun sagalee isaa meeshaalee biroo waliin akka wal hin rukunne gargaara.

2.4. FAKKEENYA — MANA AMANTAA

Mana amantaa keessatti:

Channel 1 → Vocal 1
Channel 2 → Vocal 2
Channel 3 → Keyboard
Channel 4 → Guitar
Channel 5 → Microphone
Channel 6 → Drum

Channel hundumaa irraa signal dhufu:

CHANNELS
↓
MAIN MIX
↓
AMPLIFIER
↓
MAIN SPEAKERS

Akkasumas AUX fayyadamuun:

MIXER
↓
AUX SEND
↓
MONITOR AMPLIFIER
↓
STAGE MONITOR

gochuun artistoonni ykn worship team sagalee isaanii stage irratti dhaga'uu danda'u.

3. ብቅቢ MIXER IRRATTI AKKAMITTI HOJJETU?

Analog Mixer irratti channel tokko keessatti signal yeroo baay'ee karaa kana hordofa:

3.1. INPUT

Input jechuun bakka signal audio ittiin mixer keessa seenuudha.

Microphone yeroo baay'ee XLR input fayyadama.

Meeshaalee line-level immoo input isaanii irratti hundaa'uun walqabsiifamu.

3.2. GAIN / PREAMP

Gain ykn Trim input signal qindeessa.

Microphone irraa signal baay'ee xiqqaa dhufu gara sadarkaa mixer keessatti ittiin hojjetamuutti olkaasa.

Gain baay'ee gadi yoo ta'e signal dadhabaa ta'uu danda'a.

Gain baay'ee ol yoo ta'e clipping fi distortion uumamuu danda'a.

3.3. HPF

HPF jechuun High-Pass Filter.

Frequency gadi aanaa hin barbaachifne hir'isuuf fayyada.

Fakkeenyaaf vocal microphone irratti:

- Rumble
- Handling noise
- Low-frequency vibration

hir'isuuf gargaaruu danda'a.

3.4. EQ

EQ'n frequency gara garaa sirreessa.

LOW → Bass

MID → Midrange

HIGH → Treble

Sound Engineer EQ fayyadamuun sagalee qulqulleessuu fi madaalawaa gochuu danda'a.

3.5. AUX

AUX jechuun Auxiliary Send.

AUX'n signal channel tokko irraa gara bakka biraatti erguuf fayyada.

Fakkeenya:

VOCAL
↓
AUX SEND
↓
MONITOR
↓
SINGER

Kana jechuun singer tokko sagalee ofii isaa monitor keessatti dhaga'uu danda'a.

3.6. FX

FX Send'n signal gara effects processor ykn internal effectitti erguuf fayyada.

Fakkeenya:

VOCAL
↓
FX SEND
↓
REVERB
↓
FX RETURN
↓
MAIN MIX

Reverb'n sagalee vocal irratti room ykn space fakkaatu uumu danda'a.

3.7. PAN

Pan jechuun signal gara Left fi Right stereo keessatti qooduudha.

LEFT ← CENTER → RIGHT

Pan gara center yoo ta'e:

Left fi Right irratti signal walitti dhiyaata.

Pan gara Left yoo geesse:

Signal gara Left caalaatti deema.

Pan gara Right yoo geesse:

Signal gara Right caalaatti deema.

3.8. FADER

Fader'n level channel tokkoo mix keessatti to'ata.

Fader olkaasuun channel level dabala.

Fader gadi buusuun channel level hir'isa.

Garuu fader'n gain bakka hin bu'u.

Gain input signal qindeessa.

Fader mix keessatti level channel qindeessa.

3.9. MAIN MIX

Channel gara garaa yeroo walitti makaman Main Mix uumu.

Fakkeenya:

VOCAL
+
KEYBOARD
+
GUITAR
+
DRUM
↓
MAIN MIX
↓
AMPLIFIER
↓
MAIN SPEAKER

4. ⚠ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — GAIN BAAY'EE OLKAASUU

Gain baay'ee ol yoo ta'e:

- Clipping
- Distortion
- Harsh sound
- Overload

uumamuu danda'a.

Furmaata:

Gain suuta olkaasi, meter ilaali, gurraan dhaggeeffadhu.

DOGOGGORA 2 — FADER FI GAIN WALITTI MAKUU

Gain input signal qindeessa.

Fader channel level mix keessatti qindeessa.

Lamaan isaanii hojii adda qabu.

DOGOGGORA 3 — EQ OSOO HIN DHAGGEEFFATIN JIJJIIRUU

Knob hunda olkaasuun sagalee gaarii hin godhu.

Dura dhaggeeffadhu.

Rakkoo adda baasi.

Sana booda EQ sirreessi.

DOGOGGORA 4 — FEEDBACK IRRAA OF EEGUU DHISUU

Microphone speaker fuuldura kaa'uun feedback fiduu danda'a.

Microphone fi speaker bakka sirrii kaa'i.

Gain sirriitti qindeessi.

Monitor level to'achi.

DOGOGGORA 5 — SIGNAL FLOW DAGATUU

Rakkoo yeroo uumamu meeshaa tokko qofa himatanii hin jalqabin.

Tartiiba kana hordofi:

SOURCE

↓

CABLE

↓

INPUT

↓

GAIN

↓

CHANNEL

↓

MAIN MIX

↓

AMPLIFIER

↓

SPEAKER

DOGOGGORA 6 — SAGALEE GUDDAA QOFA BARBAADUU

Sagalee guddaan yeroo hunda sagalee gaarii miti.

Sound Engineer tokko:

- Clarity
- Balance
- Headroom
- Low noise
- Controlled volume

eeguu qaba.

DOGOGGORA 7 — MEESHAA OSOO HIN HUBATIN POWER BANUU

Connection hunda dura sakatta'i.

Microphone, mixer, amplifier fi speaker sirriitti walqabsiifamuu isaanii mirkaneessi.

Yeroo system cufnus tartiiba sirrii hordofi.

YAADA IJOO

SOUND ENGINEERING jechuun:

SAGALEE QABUU

↓

SIGNAL QINDEESSUU

↓

MIX GOCHUU

↓

POWER KENNAA

↓

SPEAKER IRRAA BAASUU

↓

DHAGGEEFFATAA GEEJJUU

Sound Engineer gaarii jechuun nama mixer button hunda beekuu qofa miti.

Inni:

SAGALEE DHAGGEEFFATA

→ RAKKOO ADDA BAASA

→ SABABA HUBATA

→ MIXER IRRATTI SIRREESSA

→ SAGALEE QULQULLUU UUMA.

BOQONNAA 2 — BU'UURA SAGALEE (FUNDAMENTALS OF SOUND)

1. IBSA BAL'AA

1.1. SOUND MAAL Jechuudha?

Sound jechuun vibration ykn raafama wanta tokko irraa uumamuun medium, yeroo baay'ee qilleensa, keessa darbuun gurra namaa ga'uudha.

Yeroo nama tokko dubbatu ykn sirbu, vocal cord isaa ni raafama. Raafamni kun qilleensa naannoo isaa jiru sochoosa. Sochiin kun gurra namaa ga'uun sammuu keessatti akka sagaleetti hubatama.

Sirni bu'uuraa:

SOURCE

↓

VIBRATION

↓

AIR

↓

EAR

↓

BRAIN

1.2. VIBRATION MAAL Jechuudha?

Vibration jechuun sochii ykn raafama irra deddeebi'amuudha.

Meeshaalee muuziqaa hedduun vibration uumuun sagalee maddisiisu.

Fakkeenya:

- Vocal cord namaa
- String guitar
- Drum head
- Speaker cone
- Keyboard speaker

Yeroo vibration uumamu, qilleensi naannoo isaa jiru socho'a.

Sound Engineering keessatti vibration hubachuun baay'ee barbaachisaa dha, sababni isaa sagaleen hundi vibration irraa jalqaba.

1.3. FREQUENCY MAAL Jechuudha?

Frequency jechuun vibration tokko sekondii tokko keessatti yeroo meeqa akka raawwatamuudha.

Frequency'n Hertz (Hz) tiin safarama.

1 Hz = vibration 1 sekondii keessatti.

100 Hz = vibration 100 sekondii keessatti.

1,000 Hz = 1 kHz.

Frequency'n pitch sagalee irratti dhiibbaa qaba.

Frequency gadi aanaa:

→ Sagalee gadi fagoo / Bass

Frequency ol'aanaa:

→ Sagalee ol fagoo / Treble

1.4. LOW, MID FI HIGH FREQUENCY

LOW FREQUENCY

Low frequency jechuun frequency gadi aanaa.

Inni sagalee keessatti:

- Bass
- Weight
- Body
- Warmth

uuma.

Meeshaalee low frequency hedduu qaban:

- Bass guitar
- Kick drum
- Subwoofer
- Bass keyboard

MID FREQUENCY

Mid frequency jechuun frequency giddugaleessaa.

Vocal fi meeshaalee muuziqaa hedduu keessatti gahee guddaa qaba.

Fakkeenya:

- Vocal
- Guitar
- Keyboard

- Snare
- Speech

HIGH FREQUENCY

High frequency jechuun frequency ol'aanaa.

Inni:

- Brightness
- Clarity
- Detail
- Air

sagalee keessatti dabala.

Fakkeenya:

- Cymbal
- Hi-hat
- Treble
- Sagalee "S"
- Sagalee "T"

1.5. PITCH FI FREQUENCY

Pitch jechuun akka sagaleen gurra keenya keessatti gadi ykn ol fakkaatuudha.

Frequency fi Pitch walitti dhufeenya qabu.

Frequency olka'aa:

→ Pitch olka'aa.

Frequency gadi bu'aa:

→ Pitch gadi fagoo.

Fakkeenya:

Bass guitar → pitch gadi fagoo.

Cymbal → pitch ol fakkaatu.

1.6. AMPLITUDE

Amplitude jechuun cimina ykn guddina vibration signal tokkoo agarsiisa.

Amplitude dabaluun sagaleen dhagahamu akka caalaatti jabaatu gochuu danda'a.

Amplitude hir'isuun sagalee akka laafu godha.

Garuu amplitude fi frequency waan tokko miti.

Frequency → sagaleen gadi ykn ol fakkaachuu isaa irratti dhiibbaa qaba.

Amplitude → cimina sagalee irratti dhiibbaa qaba.

1.7. WAVELENGTH

Wavelength jechuun dheerina wave tokkoo irraa point tokko gara point itti aanu wal fakkaatutti jiruudha.

Frequency fi wavelength walitti dhufeenya qabu.

Frequency yoo dabale, wavelength yeroo baay'ee ni gabaabbata.

Frequency yoo hir'ate, wavelength ni dheerata.

2. FAKKEENYA SAGALEE

2.1. FAKKEENYA LOW FREQUENCY

Kick drum tokko yeroo rukutamu:

“BOOM”

fakkaatu dhaga'amuu danda'a.

Kun low frequency keessaa humna qabaachuu danda'a.

Bass guitar:

“DUUM”

fakkaatu sagalee gadi fagoo qaba.

Subwoofer immoo low frequency gadi fagoo uummata.

2.2. FAKKEENYA MID FREQUENCY

Vocal namaa:

“LA LA LA...”

Mid frequency keessatti humna guddaa qabaachuu danda'a.

Guitar:

“GIRR... GIRR...”

Keyboard:

“DO... RE... MI...”

Meeshaaleen kun hundi midrange keessatti odeeffannoo sagalee hedduu qabu.

2.3. FAKKEENYA HIGH FREQUENCY

Cymbal:

“TSHHHH”

Hi-hat:

“TSS TSS”

Sagalee “S” fi “T”:

“SAGALEE”

“TOLA”

High frequency keessatti odeeffannoo qabu.

2.4. FAKKEENYA AMPLITUDE

Nama tokko suuta yoo dubbate:

“akkam...”

Sagaleen laafaa ta’a.

Nama tokko sagalee isaa olkaasee yoo dubbate:

“AKKAM!”

Sagaleen jabaata.

Kun cimina sagalee ykn amplitude irratti jijjiirama agarsiisa.

2.5. FAKKEENYA FREQUENCY

Sagalee gadi fagoo:

“UUUU...”

Sagalee ol’aanaa:

“III...”

Lamaan isaanii cimina wal fakkaataa qabaachuu danda’u, garuu pitch isaanii adda.

Kun frequency fi amplitude garaagarummaa isaanii hubachuuf fakkeenya gaarii dha.

3. ብቅጥጥ MIXER IRRATTI AKKAMITTI HOJJETU?

3.1. FREQUENCY FI EQ

Analog Mixer keessatti Low, Mid fi High yeroo baay'ee EQ section keessatti argamu.

LOW → Bass

MID → Midrange

HIGH → Treble

3.2. LOW KNOB

Low knob'n frequency gadi aanaa to'ata.

Low olkaasuun:

→ Bass dabala.

Low gadi buusuun:

→ Bass hir'isa.

Fakkeenya vocal tokko baay'ee boomy yoo ta'e, Low xiqqoo hir'isuun ni danda'ama.

3.3. MID KNOB

Mid knob'n frequency giddugaleessaa to'ata.

Vocal keessatti Mid gahee guddaa qaba.

Mid baay'ee yoo ta'e:

→ Sagaleen dhiphaa ykn boxy ta'uu danda'a.

Mid xiqqoo hir'isuun ykn sirreessuun sagalee qulqulleessuu danda'a.

3.4. HIGH KNOB

High knob'n frequency ol'aanaa to'ata.

High olkaasuun:

→ Brightness fi detail dabala.

High baay'ee ol yoo ta'e:

→ Sagaleen qaraa ykn harsh ta'uu danda'a.

3.5. FREQUENCY FI GAIN

Mixer keessatti Gain input signal qindeessa.

EQ immoo frequency qindeessa.

Kanaaf:

GAIN

→ Signal level

EQ

→ Frequency balance

3.6. METER IRRATTI

Mixer tokko irratti meter ykn LED level signal agarsiisuu danda'a.

Sound Engineer meter ilaaluun:

- Signal jiraachuu
- Signal baay'ee xiqqaa ta'uu
- Signal olka'uu
- Clipping/overload

adda baasuu danda'a.

Garuu meter qofa irratti hin hirkatu.

Gurraanis dhaggeeffachuu qaba.

4. DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — FREQUENCY FI VOLUME WALITTI MAKUU

Frequency jechuun volume miti.

Frequency → Pitch/tonal content

Amplitude/Level → Cimina signal

DOGOGGORA 2 — LOW HUNDAA OLKAASUU

Low baay'ee dabaluun:

- Boomy
- Muddy
- Feedback

fiduu danda'a.

Low rakkoo qabu yoo jiraate, dura xiqqoo cut yaali.

DOGOGGORA 3 — HIGH HUNDAA OLKAASUU

High baay'ee dabaluun:

- Harshness
- Sagalee qaraa
- Gurra dhiphisuu

fiduu danda'a.

DOGOGGORA 4 — EQ OSOO DHAGGEEFFATIN JIJJIIRUU

Sound Engineer tokko:

DHAGGEEFFATA

↓

RAKKOO ADDA BAASA

↓

FREQUENCY BARBAADA

↓

CUT/BOOST XIQQOO GODHA

↓

DEEBI'II DHAGGEEFFATA

DOGOGGORA 5 — MEESHALEE HUNDAA EQ TOKKOON QINDEESSUU

Vocal, Guitar, Keyboard, Drum fi Bass hundi frequency distribution adda addaa qabu.

Kanaaf channel hundumaaf setting tokko fayyadamuun sirrii miti.

DOGOGGORA 6 — SAGALEE GUDDAA QOFA BARBAADUU

Sagalee guddaan sagalee qulqulluu jechuu miti.

Sound Engineer tokko:

- Balance
- Clarity
- Controlled level
- Low noise
- Headroom

eeguu qaba.

BOQONNAA 3 — SIGNAL FLOW FI ANALOG MIXER

1. IBSA BAL'AA

3.1. SIGNAL FLOW MAAL JECUUUDHA?

Signal Flow jechuun karaa signal audio tokko irraa ka'ee gara output isaatti ittiin darbuudha.

Sound Engineer tokko Signal Flow yoo sirriitti hubate, rakkoo sagalee bakka rakkoon itti uumame adda baasuu danda'a.

Sirni bu'uuraa:

SOURCE
↓
MICROPHONE
↓
CABLE
↓
MIXER INPUT
↓
GAIN / PREAMP
↓
EQ
↓
FADER
↓
MAIN MIX
↓
AMPLIFIER
↓
SPEAKER

3.2. SOURCE

Source jechuun bakka sagaleen irraa madduudha.

Fakkeenya:

- Vocal
- Keyboard
- Guitar
- Bass
- Drum
- Laptop
- Media Player

Source irraa signal audio jalqaba.

3.3. MICROPHONE

Microphone sagalee namaa ykn sagalee meeshaa tokkoo gara electrical audio signalitti jijjiira.

Microphone gosa adda addaa qaba.

Gosoota beekamoo:

- Dynamic Microphone
- Condenser Microphone

Dynamic microphone live performance keessatti baay'ee fayyada.

Condenser microphone sensitivity ol'aanaa qabaachuu danda'a; gosoonni isaa tokko tokko phantom power barbaadu.

3.4. CABLE

Cable signal source irraa gara mixeritti geessa.

Live Sound keessatti XLR cable microphonef baay'ee fayyada.

Cable qulqulluu fi connection sirrii ta'uun noise fi signal loss hir'isuuf gargaara.

3.5. MIXER

Mixer jechuun signal audio gara garaa fudhatee:

- Walitti makuuf
- Level to'achuuf
- EQ sirreessuuf
- Monitor erguuf
- Effects dabaluuf
- Main output qopheessuuf

kan gargaaruudha.

Analog Mixer keessatti channel tokkoon tokkoon isaa control mataa isaa qaba.

3.6. CHANNEL

Channel jechuun karaa signal tokko ittiin mixer keessa seenu fi ittiin qindaa'uudha.

Fakkeenya:

CHANNEL 1 → Vocal
CHANNEL 2 → Vocal
CHANNEL 3 → Keyboard
CHANNEL 4 → Guitar
CHANNEL 5 → Bass
CHANNEL 6 → Drum

3.7. CHANNEL STRIP

Channel Strip jechuun control channel tokko keessatti argaman walitti qabee ilaalamuudha.

Channel tokko keessatti argamuu danda'u:

- Input
- Gain
- HPF
- EQ
- AUX
- FX
- Pan
- Mute
- Fader

3.8. GAIN / PREAMP

Gain ykn Trim input signal qindeessa.

Microphone irraa signal xiqqaa dhufu gara sadarkaa hojii mixeritti olkaasa.

Gain baay'ee gadi yoo ta'e:

→ Signal dadhabaa.

Gain baay'ee ol yoo ta'e:

→ Clipping/Distortion.

3.9. EQ

EQ frequency gara garaa sirreessa.

LOW → Bass

MID → Midrange

HIGH → Treble

EQ'n sagalee qulqulleessuu fi meeshaalee adda addaa wal keessaa baasuu keessatti gahee qaba.

3.10. AUX

AUX jechuun Auxiliary Send.

Signal channel tokko irraa gara output ykn monitor biraatti erguuf fayyada.

Fakkeenya:

VOCAL
↓
AUX SEND
↓
MONITOR
↓
SINGER

3.11. FX

FX Send signal gara Reverb, Delay ykn effect biraatti erguuf fayyada.

Fakkeenya:

VOCAL
↓
FX SEND
↓
REVERB
↓
RETURN
↓
MAIN MIX

3.12. PAN

Pan signal gara Left fi Right stereo keessatti qooda.

LEFT ← CENTER → RIGHT

Pan center:

→ L fi R irratti walqixa.

Pan Left:

→ Gara Left caalaatti.

Pan Right:

→ Gara Right caalaatti.

3.13. MUTE

Mute channel tokko yeroo muraasaaf output irraa cufa.

Fakkeenya:

Microphone tokko yeroo ammaa hin barbaachifne yoo ta'e:

MUTE → ON

gochuun signal isaa Main Mix irraa dhorkuu dandeenya.

3.14. FADER

Fader channel level mix keessatti to'ata.

Fader olkaasuun:

→ Channel level dabala.

Fader gadi buusuun:

→ Channel level hir'isa.

Fader'n gain bakka hin bu'u.

3.15. MAIN MIX

Channel gara garaa walitti makamuun Main Mix uuma.

Fakkeenya:

VOCAL

+

KEYBOARD

+

GUITAR

+

BASS

+

DRUM

↓

MAIN MIX

↓

AMPLIFIER

↓

MAIN SPEAKER

2. 🎧 FAKKEENYA SAGALEE

2.1. FAKKEENYA VOCAL

Nama tokko microphone irratti sirbaa jira.

Microphone → XLR → Channel 1 → Gain → EQ → Fader → Main Mix → Amplifier → Speaker

Sound Engineer channel 1 irratti vocal qindeessa.

2.2. FAKKEENYA KEYBOARD

Keyboard → Cable/DI → Channel 3 → Gain → EQ → Fader → Main Mix → Speaker

Keyboard sagalee isaa channel mataa isaa qaba.

2.3. FAKKEENYA MONITOR

Vocal → Channel 1 → AUX Send → Monitor → Singer

Singer sagalee ofii isaa stage irratti dhaga’a.

2.4. FAKKEENYA FEEDBACK

Microphone

↓

Mixer

↓

Amplifier

↓

Speaker

↓

Sagalee gara Microphone deebi’a

↓

Feedback

Feedback yeroo baay’ee sagalee “iiiiii”, “eeeeee” ykn sagalee qaraa fakkaatuun dhaga’amuu danda’a.

3. □ MIXER IRRATTI AKKAMITTI HOJJETU?

Signal tokko mixer keessa yeroo darbu:

INPUT

↓

GAIN

↓

HPF

↓

EQ

↓

AUX/FX

↓

PAN

↓

FADER

↓

MAIN MIX

Sound Engineer tokko tartiiba kana hubachuun channel tokko qindeessa.

3.1. INPUT QINDEESSUU

Microphone XLR cable fayyadamuun input channelitti galchi.

Connection sirrii ta'uu isaa mirkaneessi.

3.2. GAIN QINDEESSUU

Gain gadi aanaa irraa jalqabi.

Nama tokko akka idilee dubbatu ykn sirbu godhi.

Gain suuta olkaasi.

Meter ilaali.

Clipping irraa of eegi.

3.3. EQ QINDEESSUU

Dura EQ neutral irraa jalqabi.

Sagalee dhaggeeffadhu.

Rakkoo yoo jiraate:

- Low
- Mid
- High

keessaa bakka rakkoon jiru barbaadi.

Jijjiirama xiqqoo godhi.

3.4. AUX QINDEESSUU

Monitor barbaadde yoo ta'e:

Channel → AUX Send → Monitor

AUX level suuta qindeessi.

Monitor fi main speaker walitti dhiyeessuun feedback fiduu danda'a.

3.5. FADER QINDEESSUU

Gain sirrii erga argatte booda fader channel qindeessi.

Channel hundumaa walitti makuun:

Vocal
+
Keyboard
+
Guitar
+
Drum

balance godhi.

3.6. MAIN MIX

Main Mix level qindeessi.

Main output irraa signal gara amplifieritti darba.

Amplifier irraa gara speakeritti darba.

4. △□ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1:

Signal Flow beekuu dhiisuu.

Furmaata:

Signal eessaa akka dhufu fi eessa akka deemu yeroo hunda hordofi.

DOGOGGORA 2:

Gain baay'ee olkaasuu.

Furmaata:

Gain suuta qindeessi; clipping irraa of eegi.

DOGOGGORA 3:

Fader qofa fayyadamuun signal sirreessuu.

Furmaata:

Dura input gain sirreessi, booda fader qindeessi.

DOGOGGORA 4:

EQ button hunda yeroo tokkoon jijjiiruu.

Furmaata:

Rakkoo tokko tokkoon adda baasi; jijjiirama xiqqoo godhi.

DOGOGGORA 5:

AUX level baay'ee olkaasuu.

Furmaata:

Monitor level to'achi; feedback yoo dhaga'ame saffisaan level gadi buusi.

DOGOGGORA 6:

Pan osoo hin hubatin channel hunda Left ykn Rightitti qooduu.

Furmaata:

Jalqaba center irraa eegali; stereo image barbaachisaa ta'e qofa qindeessi.

DOGOGGORA 7:

Mute button dagachuu.

Furmaata:

Sagalee channel tokko hin dhageenye yoo ta'e Mute ilaali.

DOGOGGORA 8:

Main Mix fi channel fader walitti makuu.

Furmaata:

Channel level fi overall Main Mix level adda addaa ta'uu hubadhu.

YAADA IJOO

SIGNAL FLOW HUBADHU:

SOURCE

↓

INPUT

↓

GAIN

↓

EQ

↓

AUX/FX

↓

FADER

↓

MAIN MIX



AMPLIFIER



SPEAKER

Signal Flow sirriitti beekuun Sound Engineer tokkoof bu'uura rakkoo sagalee adda baasuu fi sirreessuu ti.

BOQONNAA 4 — GAIN, PREAMP FI SIGNAL LEVEL

1. IBSA BAL'AA

4.1. PREAMP MAAL Jechuudha?

Preamp jechuun “Preamplifier” jechuudha.

Microphone irraa dhufu signal baay'ee xiqqaa ta'uu danda'a. Signal kun mixer keessatti sirnaan hojjechuuf sadarkaa gahaa qabaachuu qaba.

Preamp'n signal microphone sana gara sadarkaa hojii mixeritti olkaasa.

Sirni isaa:

MICROPHONE
↓
WEAK AUDIO SIGNAL
↓
PREAMP
↓
USABLE SIGNAL
↓
MIXER

4.2. GAIN MAAL Jechuudha?

Gain jechuun input signal tokko gara mixeritti seenu sadarkaa isaa qindeessuudha.

Analog mixer irratti Gain ykn Trim knob yeroo baay'ee channel input irratti argama.

Gain'n:

- Input signal olkaasa.
- Signal dadhabaa ta'e cimsuuf gargaara.
- Noise fi signal ratio irratti dhiibbaa qaba.
- Headroom eeguu keessatti gahee qaba.

4.3. GAIN FI VOLUME GARAAGARUMMAA

Gain fi volume waan tokko miti.

GAIN:

Input signal qopheessa.

VOLUME/FADER:

Signal erga channel keessa darbuun booda level isaa mix keessatti to'ata.

Fakkeenya:

Microphone



GAIN



EQ



FADER



MAIN MIX

4.4. SIGNAL LEVEL

Signal Level jechuun cimina signal audio tokkoo system keessa jiruudha.

Signal baay'ee xiqqaa yoo ta'e:

→ Noise waliin walitti makamu danda'a.

Signal baay'ee guddaa yoo ta'e:

→ Clipping ykn overload uumamuu danda'a.

Kanaaf Sound Engineer signal level giddugaleessa sirrii keessatti to'achuu qaba.

4.5. HEADROOM

Headroom jechuun signal hojii fi daangaa ol'aanaa system tokkoo gidduu jiru space ykn margin dha.

Headroom gahaa qabaachuun signal tasaa olka'uuf bakka dhiisa.

Fakkeenya:

Nama tokko sirbaa jira.

Sagalee isaa yeroo baay'ee normal dha.

Tasuma sagalee isaa olkaase.

Yoo headroom gahaa jiraate, system'n signal sana qabachuu danda'a.

Yoo headroom hin jirre:

→ Clipping uumamuu danda'a.

4.6. CLIPPING

Clipping jechuun signal daangaa system tokkoo caalee waveform isaa jallachuu ykn “cituu” dha.

Sagaleen clipping qabu:

- Distorted
- Qaraa
- Jabaataa
- Unpleasant

ta’uu danda’a.

Clipping yeroo baay’ee gain ykn signal level baay’ee olkaasuu irraa uumama.

2. 🎧 FAKKEENYA SAGALEE

2.1. GAIN XIQQAA

Microphone
↓
Signal baay’ee xiqqaa
↓
Mixer
↓
Fader baay’ee ol
↓
Noise

Yoo gain input xiqqaa ta’e, Sound Engineer fader baay’ee olkaasuun signal waliin noise illee olkaasuu danda’a.

2.2. GAIN BAAY’EE OL

Microphone
↓
Gain baay’ee ol
↓
Input overload
↓
Clipping
↓
Distortion

Sagaleen qulqulluu ta’uu dhiisa.

2.3. GAIN SIRRII

Microphone
↓
Gain sirrii

↓
Signal qulqulluu
↓
EQ
↓
Fader
↓
Main Mix

Kun signal qindaa'aa fi headroom ga'aa qabaachuuf karaa gaarii dha.

2.4. VOCAL FAKKEENYA

Nama tokko microphone irratti sirbaa jira.

Yoo Gain baay'ee gadi ta'e:

Sagaleen laafaa ta'a.

Yoo Gain baay'ee ol ta'e:

Sagaleen cituu ykn distortion qabaachuu danda'a.

Yoo Gain sirrii ta'e:

Sagaleen qulqulluu fi to'atamaa ta'a.

3. ተባብሮ ማክሰር ላይ ማድረግ?

3.1. GAIN QINDEESSUU

Tartiiba sirrii:

1. Channel filadhu.
2. Fader channel gadi buusi ykn setting sirrii irraa jalqabi.
3. Gain gadi aanaa irraa jalqabi.
4. Nama tokko akka normal dubbatu ykn sirbu godhi.
5. Gain suuta olkaasi.
6. Meter/Peak LED ilaali.
7. Signal qulqulluu ta'uu isaa dhaggeeffadhu.
8. Clipping irraa of eegi.

3.2. METER

Mixer tokko irratti meter ykn Peak LED jiraachuu danda'a.

Meter'n signal level agarsiisa.

Yoo signal baay'ee xiqqaa ta'e:

→ Level gadi.

Yoo signal hojii gaarii keessa jiraate:

→ Level sirrii.

Yoo signal daangaa ga'e:

→ Peak/clipping irraa of eegi.

Lakkoofsi sirrii mixer hundumaa irratti walqixa miti; kanaaf mallattoo fi manual mixer sanaa hubachuun barbaachisaa dha.

3.3. GAIN FI FADER WAL BIRA QABUU

Fakkeenya:

Gain = input signal qopheessa.

Fader = mix keessatti channel level qindeessa.

Kanaaf:

GAIN → INPUT CONTROL

FADER → MIX CONTROL

3.4. GAIN STAGING

Gain Staging jechuun signal system keessa bakka tokko irraa gara bakka itti aanu darbu keessatti level isaa sirriitti qindeessuudha.

Fakkeenya:

Microphone

↓

Preamp

↓

EQ

↓

AUX

↓

Fader

↓

Main Mix

↓

Amplifier

Bakka hundatti signal baay'ee xiqqaa ykn baay'ee guddaa akka hin taane to'achuun barbaachisaa dha.

3.5. PRAKTIKAALAA VOCAL

Microphone tokko channel 1 irratti walqabsiisi.

Gain gadi buusi.

Nama tokko sirba ykn dubbii normal godhi.

Gain suuta olkaasi.

Meter ilaali.

Sagalee dhaggeeffadhu.

Yoo distortion dhaga'ame gain xiqqeessi.

Yoo signal dadhabaa ta'e gain xiqqoo dabali.

Ergasii fader qindeessi.

4. △□ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — GAIN FI FADER WAL FAKKEESSUU

Gain input signal qindeessa.

Fader mix level qindeessa.

DOGOGGORA 2 — FADER BAAY'EE OLKAASUU

Gain gadi ta'e fader baay'ee olkaasuun noise dabalu danda'a.

Furmaata:

Dura gain sirreessi.

DOGOGGORA 3 — GAIN BAAY'EE OLKAASUU

Gain baay'ee olkaasuun clipping fiduu danda'a.

Furmaata:

Meter fi gurra lamaan fayyadami.

DOGOGGORA 4 — HEADROOM DAGATUU

Signal yeroo tasaa olka'uuf bakka hin dhiisuun clipping fiduu danda'a.

Furmaata:

Headroom gahaa eegi.

DOGOGGORA 5 — METER QOFA ILAALUU

Meter gaarii fakkaatus sagaleen gurra namaa irratti harsh ykn distorted ta'uu danda'a.

Furmaata:

Meter + Gurra = murtii Sound Engineer.

DOGOGGORA 6 — CHANNEL HUNDAA GAIN TOKKOON QINDEESSUU

Microphone, keyboard, guitar fi drum hundi output adda addaa qabu.

Kanaaf channel tokkoon tokkoon isaa dhuunfaan qindaa'uu qaba.

YAADA IJOO

GAIN = INPUT SIGNAL QINDEESSUU

FADER = CHANNEL LEVEL MIX KEESSATTI TO'ACHUU

HEADROOM = BAKKA SIGNAL TASAA OLKA'UUF HAFU

CLIPPING = SIGNAL DAANGAA CAALUU

Sound Engineer gaarii tokko gain sirriitti qindeessuun dura sagalee dhaggeeffata; sana booda EQ, fader fi main mix irratti hojii itti fufa.

BOQONNAA 5 — EQUALIZATION (EQ)

1. IBSA BAL'AA

5.1. EQ MAAL Jechuudha?

EQ jechuun Equalization jechuudha.

Equalization jechuun frequency gara garaa sagalee tokko keessa jiran sirreessuu, hir'isuu ykn dabaluudha.

Sound Engineer EQ fayyadamuun:

- Sagalee qulqulleessa.
- Sagalee madaala.
- Frequency hin barbaachifne hir'isa.
- Meeshaalee gara garaa wal irraa adda baasa.
- Vocal akka ifa ta'u gargaara.
- Feedback to'achuuf gargaara.

5.2. FREQUENCY RANGE

Sagaleen namaa fi meeshaaleen muuziqaa frequency gara garaa qabu.

Akka salphaatti:

LOW

→ Bass

MID

→ Body / Presence

HIGH

→ Brightness / Detail

5.3. LOW FREQUENCY

Low frequency'n sagalee gadi fagoo fi ulfaataa uuma.

Fakkeenya:

- Kick
- Bass guitar
- Subwoofer
- Low vocal
- Keyboard bass

Low baay'inaan yoo jiraate:

- Boomy
- Muddy
- Heavy

ta'uu danda'a.

5.4. MID FREQUENCY

Mid frequency'n sagalee keessatti odeeffannoo guddaa qaba.

Vocal, guitar, keyboard fi instruments hedduu keessatti mid gahee guddaa qaba.

Mid baay'inaan yoo jiraate:

- Boxy
- Nasal
- Muddy

ta'uu danda'a.

5.5. HIGH FREQUENCY

High frequency'n:

- Brightness
- Clarity
- Detail
- Air

dabala.

High baay'inaan yoo jiraate:

- Harsh
- Qara
- Gurra namaa dhiphisa.

5.6. CUT FI BOOST

BOOST jechuun frequency tokko dabaluudha.

CUT jechuun frequency tokko hir'isuudha.

Fakkeenya:

LOW +3 dB

→ Low xiqqoo dabale.

HIGH -3 dB

→ High xiqqoo hir'ise.

Sound Engineer yeroo baay'ee rakkoo frequency tokko sirreessuuf CUT xiqqoo fayyadamuun bu'aa gaarii argachuu danda'a.

Garuu boost yeroo barbaachisaa ta'etti qofa fayyadamuu qaba.

5.7. PARAMETRIC EQ

EQ professional tokko keessatti:

- Frequency
- Gain
- Q

to'achuun ni danda'ama.

Frequency → bakka frequency filatu.

Gain → dabala ykn hir'isa.

Q → bal'ina ykn dhiphachuu frequency range to'ata.

Analog mixer salphaa keessatti EQ'n Low, Mid fi High qofa qabaachuu danda'a.

Mixer tokko tokko keessatti sweepable Mid ykn semi-parametric EQ jiraachuu danda'a.

2. 🎧 FAKKEENYA SAGALEE

5.8. VOCAL

Vocal tokko:

“BOOMY”

yoo dhaga'ame, Low frequency baay'achuu danda'a.

Low xiqqoo hir'isuun sagalee qulqulleessuu danda'a.

Vocal:

“BOXY”

yoo fakkaate, Mid keessaa frequency rakkoo qabu jiraachuu danda'a.

Mid xiqqoo sirreessuun gargaaruu danda'a.

Vocal:

“QARAA”

yoo ta'e, High ykn upper-mid baay'achuu danda'a.

Frequency rakkoo qabu xiqqoo hir'isuun ni danda'ama.

5.9. KICK DRUM

Kick tokko:

“BOOM”

yoo barbaadde, Low frequency isaa sirriitti qindeessi.

Attack ykn definition barbaadde yoo ta'e, Mid/High range barbaachisaa ta'uu danda'a.

5.10. BASS GUITAR

Bass guitar:

“DUUM”

Sagalee isaa keessatti low frequency gahee guddaa qaba.

Garuu low baay'inaan mix guutuu haguuguu danda'a.

Kanaaf bass fi kick walitti hin rukunneef EQ fi level sirriitti qindeessi.

5.11. KEYBOARD

Keyboard keessatti Low, Mid fi High hundi jiraachuu danda'a.

Low baay'inaan:

→ Mix dhiphisa.

Mid baay'inaan:

→ Vocal waliin wal ruka.

High baay'inaan:

→ Sagalee qaraa taasisuu danda'a.

5.12. GUITAR

Guitar acoustic ykn electric:

Low → Body

Mid → Character

High → Brightness

qabaachuu danda'a.

Guitar fi vocal frequency walitti bu'uu danda'u.

EQ'n bakka isaanii mix keessatti adda baasuu gargaara.

3. ቶክሶ ማክሰር ማዘን ማዘን ማዘን?

5.13. EQ CHANNEL KEESSATTI

Analog mixer tokko irratti EQ section yeroo baay'ee:

HIGH

MID

LOW

jedhamee argama.

Mixer tokko tokko:

HIGH

MID

LOW

qofa qaba.

Kan biraa immoo:

HIGH

HIGH-MID

LOW-MID

LOW

qabaachuu danda'a.

5.14. EQ SIRREESSUU — TARTIIBA

1. Channel filadhu.
2. EQ neutral irraa jalqabi.
3. Sagalee dhaggeeffadhu.
4. Rakkoo adda baasi.
5. Frequency rakkoo qabu barbaadi.
6. Cut xiqqoo yaali.
7. Deebitee dhaggeeffadhu.
8. Yoo barbaachise boost xiqqoo godhi.

5.15. VOCAL EQ — FAKKEENYA

Vocal

↓

Gain
↓
HPF yoo barbaachise
↓
Low sirreessi
↓
Mid sirreessi
↓
High sirreessi
↓
Fader
↓
Main Mix

Kaayyoon EQ vocal:

- Low mud hin barbaachifne hir'isuu.
- Mid rakkoo qabu sirreessuu.
- High clarity barbaachisaa ta'e qindeessuu.

5.16. MIX BALANCE

EQ channel tokko qofa irratti ilaaluun hin ga'u.

Fakkeenya:

Vocal + Guitar

Lamaan isaanii midrange hedduu yoo qabaatan, wal haguuguu danda'a.

Sound Engineer:

- EQ
- Level
- Pan

fayyadamuun isaan adda baasuu danda'a.

5.17. FEEDBACK FI EQ

Feedback yeroo tokko frequency murtaa'e irratti baay'ee mul'achuu danda'a.

Sound Engineer frequency sana adda baasee xiqqoo cut gochuu danda'a.

Garuu feedback yeroo hunda EQ qofaan furamu hin qabu.

Microphone placement, speaker placement, gain fi monitor level illee sakatta'amu qabu.

4. ▴▢ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — KNOB HUNDAA OLKAASUU

Low + High + Mid hunda dabaluun sagalee gaarii hin godhu.

Furmaata:

Dura rakkoo adda baasi.

DOGOGGORA 2 — BOOST BAAY'EE GOCHUU

Frequency hundumaa boost gochuun:

- Muddy mix
- Harsh sound
- Feedback
- Headroom hir'achuu

fiduu danda'a.

DOGOGGORA 3 — EQ OSOO SOLO DHAGGEEFFATIN QINDEESSUU

Sagaleen solo keessatti gaarii dhaga'amu mix keessatti gaarii ta'uu dhiisuu danda'a.

EQ yeroo baay'ee mix guutuu keessatti dhaggeeffachuun sirrii dha.

DOGOGGORA 4 — FREQUENCY TOKKO QOFA IRRATTI HIRKACHUU

Lakkoofsi frequency qofa murtii hin kennu.

Microphone, room, speaker, performer fi meeshaa hundi garaagarummaa qaba.

DOGOGGORA 5 — FEEDBACK HUNDAA EQ QOFAan FURUU

Feedback yoo uumame:

- Mic position
- Speaker position
- Gain
- Monitor level
- EQ

hunda sakatta'i.

DOGOGGORA 6 — CUT FI BOOST HIKEE

CUT → Frequency hir'isa.

BOOST → Frequency dabala.

Lamaan isaanii hojii adda qabu.

DOGOGGORA 7 — EQ BAAY'EE JIJJIIRUU

Jijjiirama guddaa yeroo tokkoon gochuun sagalee uumamaa balleessuu danda'a.

Furmaata:

Jijjiirama xiqqoo godhi.

Deebitee dhaggeeffadhu.

Sana booda murteessi.

YAADA IJOO

EQ = FREQUENCY SIRREESSUU

LOW = BASS

MID = BODY / CHARACTER

HIGH = BRIGHTNESS / DETAIL

CUT = HIR'ISUU

BOOST = DABALUU

Sound Engineer gaarii EQ button tuquu qofa osoo hin taane, **rakkoo sagalee gurraan adda baasee frequency sirrii irratti tarkaanfii sirrii fudhata.**

BOQONNAA 6 — AUX SEND, MONITOR FI MONITOR MIX

1. IBSA BAL'AA

6.1. AUX SEND MAAL Jechuudha?

AUX jechuun **Auxiliary Send** jechuudha.

AUX Send jechuun signal channel tokko ykn channeloota hedduu irraa fudhatee gara output addaatti erguudha.

Main Mix tokko qofa osoo hin taane, Sound Engineer AUX fayyadamuun mix adda addaa uumuu danda'a.

Fakkeenya:

CHANNEL

↓

AUX SEND

↓

MONITOR

↓

PERFORMER

AUX'n Live Sound keessatti keessumaa:

- Stage Monitor
- In-Ear Monitor
- Effects
- External Processor
- Recording/Other Outputs

fayyadamuuf gargaaruu danda'a.

6.2. MONITOR MAAL Jechuudha?

Monitor jechuun speaker performer ykn nama stage irratti hojjetuuf sagalee isaa ykn meeshaalee barbaachisan dhageessisuudha.

Main Speaker uummataaf sagalee kenna.

Monitor immoo performer'f sagalee barbaachisaa ta'e kenna.

Fakkeenya:

MAIN SPEAKER → UUMMATA

MONITOR → ARTIST / WORSHIP TEAM

6.3. MONITOR MIX MAAL Jechuudha?

Monitor Mix jechuun mix addaa stage irratti nama tokko ykn garee tokkoof qophaa'edha.

Fakkeenya:

Singer tokko:

- Sagalee ofii isaa baay'ee
- Keyboard xiqqoo
- Guitar xiqqoo
- Drum xiqqoo

barbaaduu danda'a.

Singer biraa immoo:

- Vocal
- Keyboard
- Guitar

caalaatti barbaaduu danda'a.

Kanaaf performer hundumaaf mix tokko qofa kennuun yeroo hunda sirrii miti.

6.4. PRE-FADER FI POST-FADER

AUX Send gosa adda addaa qabaachuu danda'a.

PRE-FADER:

AUX level'n channel fader main irraa walaba ta'uu danda'a.

Kun monitor mix keessatti baay'ee faayidaa qaba.

POST-FADER:

AUX level'n channel fader main waliin walitti hidhata qabaachuu danda'a.

Kun effects keessatti yeroo baay'ee fayyada.

Garuu mixer hundi design tokko miti. Kanaaf mixer kee irratti AUX tokko Pre ykn Post ta'uu isaa mallattoo isaa ykn manual isaa irraa mirkaneessi.

6.5. MONITOR FI MAIN SPEAKER GARAAGARUMMAA

MAIN SPEAKER:

→ Uummataaf.

MONITOR:

→ Performer'f.

Main Speaker fi Monitor bakka sirrii kaa'amuu qabu.

Sababni isaa microphone fi speaker walitti dhiyaachuun feedback fiduu danda'a.

2. 🎵 FAKKEENYA SAGALEE

6.6. FAKKEENYA SINGER

Singer tokko stage irratti jira.

Microphone → Channel 1

Sound Engineer:

Channel 1

↓

AUX 1

↓

Monitor 1

↓

Singer

Monitor 1 keessatti:

Vocal → Baay'ee

Keyboard → Giddugaleessa

Guitar → Xiqqoo

ta'uu danda'a.

6.7. FAKKEENYA KEYBOARDIST

Keyboardist tokko:

Keyboard → Baay'ee

Vocal → Giddugaleessa

Drum → Xiqqoo

barbaaduu danda'a.

Kun monitor mix addaa ta'uu danda'a.

6.8. FAKKEENYA WORSHIP TEAM

Fakkeenyaaf:

AUX 1 → Vocal Monitor

AUX 2 → Keyboard Monitor

AUX 3 → Band Monitor

AUX 4 → In-Ear Monitor

Kana jechuun AUX adda addaatiin mix adda addaa uumuu danda'ama.

6.9. FAKKEENYA FEEDBACK

Microphone

↓

Monitor Speaker

↓

Sagalee gara Microphone deebi'a

↓

Feedback

Yeroo feedback:

“IIIIII...”

“EEEEEEE...”

fakkaatu dhaga'amuu danda'a.

Feedback uumamuun volume qofa irraa miti.

Position, gain, EQ, room fi speaker direction illee sababoota ta'uu danda'u.

3. □ MIXER IRRATTI AKKAMITTI HOJJETU?

6.10. AUX SEND QINDEESSUU

Fakkeenya:

Channel 1 = Vocal

AUX 1 = Vocal Monitor

Tartiiba:

CHANNEL 1

↓

AUX 1

↓

MONITOR OUTPUT



MONITOR SPEAKER

6.11. MONITOR MIX QINDEESSUU

Dura AUX output filadhu.

Channel barbaachisan AUX irratti dabali.

Fakkeenya:

Vocal → AUX 1 = ol

Keyboard → AUX 1 = giddugaleessa

Guitar → AUX 1 = xiqqoo

Drum → AUX 1 = barbaachisummaa irratti hundaa'a

Ergasii AUX Master level qindeessi.

6.12. PRE-FADER AUX

Yoo AUX tokko Pre-Fader ta'e:

Channel fader main jijjiiramus monitor mix'n isaa walaba ta'uu danda'a.

Kun performer tokkoof monitor mix addaa qopheessuuf baay'ee faayidaa qaba.

6.13. POST-FADER AUX

Yoo AUX tokko Post-Fader ta'e:

Channel fader main jijjiiramuu waliin AUX send level illee jijjiiramuu danda'a.

Effects keessatti kun faayidaa qabaachuu danda'a.

6.14. AUX MASTER

AUX channel tokkoon tokkoon isaa irraa ergamee gara AUX Masteritti walitti qabama.

Fakkeenya:

CH1 → AUX 1

CH2 → AUX 1

CH3 → AUX 1

CH4 → AUX 1



AUX 1 MASTER

↓
MONITOR

6.15. MONITOR LEVEL

Monitor level suuta olkaasi.

Performer irraa feedback ykn sagalee hin barbaachifne yoo dhaga'ame:

- AUX send
- Monitor master
- Channel level
- EQ
- Microphone position

sakatta'i.

4. △□ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — AUX HUNDAA WAL FAKKAATTI QINDEESSUU

Performer hundi waan tokko hin barbaadu.

Furmaata:

Nama tokkoon tokkoon isaa maal akka barbaadu hubadhu.

DOGOGGORA 2 — MONITOR LEVEL BAAY'EE OLKAASUU

Monitor baay'ee olkaasuun feedback fiduu danda'a.

Furmaata:

Level suuta olkaasi.

DOGOGGORA 3 — PRE FI POST WALITTI MAKUU

AUX tokko Pre ykn Post ta'uu isaa hubachuu dhabuun monitor ykn FX rakkoo fiduu danda'a.

Furmaata:

AUX mode mixer kee irratti mirkaneessi.

DOGOGGORA 4 — MAIN MIX QOFA DHAGGEEFFACHUU

Sound Engineer stage monitor mix illee dhaggeeffachuu qaba.

Furmaata:

Main Mix fi Monitor Mix lamaan isaanii sakatta'i.

DOGOGGORA 5 — MICROPHONE MONITOR FUULDURA KAAUU

Microphone speaker fuuldura yoo jiraate feedback saffisaan uumamuu danda'a.

Furmaata:

Microphone fi monitor bakka sirrii kaa'i.

DOGOGGORA 6 — AUX MASTER DAGATUU

Channel AUX level sirrii ta'us AUX Master gadi yoo ta'e monitor irraa sagaleen hin bahu.

Furmaata:

Channel AUX Send fi AUX Master lamaan isaanii ilaali.

DOGOGGORA 7 — AUX FI MAIN FADER WAL QIXA SE'UU

AUX mix fi Main Mix hojii adda qabu.

AUX → Monitor/other output

Main → Main Speaker

YAADA IJOO

AUX SEND = SIGNAL GARA OUTPUT ADDAATTI ERGUU

MONITOR = PERFORMER'F SAGALEE KENNUU

PRE-FADER = MAIN FADER IRRAA WALABA TA'UU DANDA'A

POST-FADER = MAIN FADER WALIIN WALQABATUU DANDA'A

Sound Engineer gaarii tokko performer tokkoon tokkoof **mix barbaachisaa ta'e** qopheessa; monitor level fi feedback sirriitti to'ata.

BOQONNAA 7 — MICROPHONE FI MICROPHONE TECHNIQUE

1. IBSA BAL'AA

7.1. MICROPHONE MAAL JECUUUDHA?

Microphone jechuun sagalee ykn sound waves gara electrical audio signalitti jijjiiruuf meeshaa fayyadamnuudha.

Sirni isaa:

SOUND WAVE

↓

MICROPHONE

↓

AUDIO SIGNAL

↓

CABLE

↓

MIXER

Microphone Sound Engineering keessatti bakka jalqabaa signal chain keessaa tokko qaba.

Kanaaf microphone sirriitti filachuu, kaa'uu fi fayyadamuun sagalee qulqulluu argachuuf baay'ee barbaachisaa dha.

7.2. GOSOOTA MICROPHONE

Microphone gosa adda addaa qaba.

Gosoota bu'uuraa:

- Dynamic Microphone
- Condenser Microphone
- Ribbon Microphone

Dynamic microphone live performance keessatti baay'ee fayyada.

Condenser microphone sensitivity ol'aanaa qabaachuu danda'a; yeroo baay'ee phantom power barbaada.

Ribbon microphone immoo hojii studio fi recording keessatti haala adda addaatiin fayyadamuu danda'ama.

7.3. POLAR PATTERN

Polar Pattern jechuun microphone'n sagalee kallattii kami irraa caalaatti akka qabatu agarsiisa.

Gosoota beekamoo:

- Cardioid
- Supercardioid
- Hypercardioid
- Omnidirectional
- Figure-8

Cardioid microphone fuuldura irraa sagalee caalaatti qabata; duuba irraa sagalee xiqqoo fudhata.

Kun live stage keessatti feedback hir'isuuf faayidaa qabaachuu danda'a.

7.4. PROXIMITY EFFECT

Microphone directional tokko keessatti yeroo source microphone'itti baay'ee dhiyaatu, low frequency dabaluun sagalee isaa ulfaataa gochuu danda'a.

Kana Proximity Effect jedhama.

Fakkeenya:

Microphone irraa fagoo → sagalee uumamaa

Microphone baay'ee dhihoo → bass/low dabachuu danda'a.

7.5. MICROPHONE PLACEMENT

Microphone placement jechuun microphone bakka sirrii kaa'uudha.

Placement sirrii:

- Sagalee qulqulleessa.
- Feedback hir'isa.
- Noise hir'isa.
- EQ irratti hojii hir'isa.
- Performance fooyyeessa.

Placement dogoggoraa:

- Feedback
- Muddy sound
- Unwanted noise
- Weak signal

fiduu danda'a.

2. 🎙️ FAKKEENYA SAGALEE

7.6. VOCAL MICROPHONE

Singer tokko microphone qabatee sirbaa jira.

Microphone



Vocal



XLR Cable



Mixer

Singer microphone irraa fageenya wal fakkaataa qabaachuun sagalee isaa akka tasgabbaa'u gargaara.

Microphone afaanitti maxxansuu ykn baay'ee irraa fageessuun level jijjiirama fiduu danda'a.

7.7. DRUM MICROPHONE

Kick drum:

Kick → Microphone → Mixer

Sagalee kick keessatti low frequency fi attack barbaachisaa ta'uu danda'a.

Snare:

Snare → Microphone → Mixer

Snare keessatti attack fi mid/high detail barbaachisaa ta'uu danda'a.

7.8. GUITAR AMPLIFIER

Guitar amp fuulduratti microphone kaa'uun:

Guitar Amp



Microphone



Mixer

signal argamsiisa.

Microphone speaker cone irratti bakka gara garaa kaa'uun sagalee adda addaa kennuu danda'a.

7.9. MICROPHONE FI SPEAKER

Microphone



Mixer



Amplifier



Speaker

↘

Microphone

Yoo speaker irraa sagaleen gara microphone deebi'e, feedback uumamuu danda'a.

3. ብላክ ባክላንት MIXER IRRATTI AKKAMITTI HOJJETU?

7.10. MICROPHONE CHANNEL QINDEESSUU

Microphone XLR cable fayyadamuun channel inputitti galchi.

Ergasii:

GAIN

↓

HPF

↓

EQ

↓

AUX

↓

FX

↓

FADER

↓

MAIN MIX

7.11. DYNAMIC MICROPHONE

Dynamic microphone tokko yeroo live vocal irratti fayyadamu:

Microphone → XLR → Mixer Input

Gain sirriitti qindeessi.

EQ sagalee irratti hundaa'uun sirreessi.

Fader'n level isaa qindeessi.

7.12. CONDENSER MICROPHONE

Condenser microphone tokko power barbaaduu danda'a.

Mixer tokko irratti phantom power (+48V) qabaachuu danda'a.

Garuu phantom power yeroo banuu dura:

- Cable
- Microphone

- Mixer
- Connected equipment

sirriitti mirkaneessi.

Phantom power meeshaalee tokko tokko waliin rakkoo fiduu danda'a; kanaaf yeroo hunda manual fi manufacturer instructions hordofi.

7.13. MICROPHONE LEVEL

Microphone irraa signal dhufu:

GAIN → Input level

FADER → Mix level

Kanaaf microphone level sirreessuuf gain fi fader hojii adda addaa qabu.

7.14. MICROPHONE FI MONITOR

Microphone tokko monitor speaker waliin yeroo hojii irra oolu:

Microphone

↓

Mixer

↓

AUX

↓

Monitor

Microphone fi monitor placement sirrii ta'uu qaba.

Monitor level baay'ee olkaasuun feedback fiduu danda'a.

7.15. MULTIPLE MICROPHONES

Microphone hedduu yeroo fayyadamnu:

Vocal 1 → CH1

Vocal 2 → CH2

Vocal 3 → CH3

Speech → CH4

Drum → CH5+

Channel tokkoon tokkoon isaa maqaa ykn label qabaachuun hojii Sound Engineer salphisa.

4. ለ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — MICROPHONE AFANITTI MAXXANSUU

Microphone baay'ee dhihoo qabachuun proximity effect fi plosive dabalu danda'a.

Furmaata:

Fageenya madaalawaa eegi.

DOGOGGORA 2 — MICROPHONE BAAY'EE FAGEESSUU

Microphone baay'ee fageessuun:

- Signal gadi buusu
- Room sound dabalu
- Noise dabalu

danda'a.

DOGOGGORA 3 — MICROPHONE SPEAKER FUULDURA KAAUU

Kun feedback fiduu danda'a.

Furmaata:

Microphone fi speaker kallattii sirrii irratti kaa'i.

DOGOGGORA 4 — MICROPHONE HEAD TUQUU

Microphone head ykn grille harkaan cufuun polar pattern isaa jijjiiruu fi sagalee balleessuu danda'a.

Furmaata:

Microphone body irraa qabachuun sirrii dha.

DOGOGGORA 5 — MICROPHONE HUNDAA WAL FAKKAATTI FAYYADAMUU

Microphone hundi source hundumaaf mijataa miti.

Furmaata:

Source, room fi application irratti hundaa'uun microphone filadhu.

DOGOGGORA 6 — PHANTOM POWER OSOO HIN HUBATIN BANUU

Condenser microphone tokko +48V barbaaduu danda'a.

Garuu phantom power meeshaalee tokko tokko waliin walitti bu'uu danda'a.

Furmaata:

+48V banuu dura setup guutuu mirkaneessi.

DOGOGGORA 7 — CABLE QORACHUU DHIISUU

Microphone sagalee yoo hin kennine, microphone qofa hin himatin.

Sakatta'i:

Microphone

↓

XLR Cable

↓

Mixer Input

↓

Gain

↓

Channel Mute

↓

Fader

↓

Main Mix

YAADA IJOO

MICROPHONE = SOUND → AUDIO SIGNAL

MICROPHONE FILANNAA SIRRII

+

PLACEMENT SIRRII

+

GAIN SIRRII

+

EQ SIRRII

+

MONITOR CONTROL

= SAGALEE QULQULLUU

Sound Engineer tokko microphone sirriitti fayyadamuun rakkoo hedduu osoo mixer irratti hin furin dura source irraa sirreessuu danda'a.

BOQONNAA 8 — FEEDBACK, NOISE FI HUM

1. IBSA BAL'AA

8.1. FEEDBACK MAAL Jechuudha?

Feedback jechuun sagalee speaker irraa ba'ee deebi'ee microphone'n qabamuun system keessa irra deddeebi'amee guddataa deemudha.

Sirni isaa:

MICROPHONE

↓

MIXER

↓

AMPLIFIER

↓

SPEAKER

↓

SOUND

↘

MICROPHONE

↓

MIXER

↓

AMPLIFIER

↓

SPEAKER

Yoo loop kun itti fufe, sagaleen tokko yeroo baay'ee guddataa deema.

Feedback yeroo baay'ee sagalee:

“IIIIIII...”

“EEEEEEE...”

“UUUUUU...”

fakkaatuun dhaga'amuu danda'a.

8.2. FEEDBACK MAALIIF UUMAMA?

Sababoota ijoo:

- Microphone speakeritti baay'ee dhihoo ta'uu.
- Microphone speaker fuuldura jiraachuu.
- Monitor level baay'ee olka'uu.
- Gain baay'ee olka'uu.
- Frequency tokko system keessatti baay'achuu.

- Room acoustics rakkoo qabaachuu.
- Microphone polar pattern sirrii hin taane fayyadamuu.

8.3. NOISE MAAL Jechuudha?

Noise jechuun sagalee hin barbaachifne audio signal keessa seenuudha.

Fakkeenya:

- Hiss
- Hum
- Buzz
- Crackle
- Electrical interference
- Cable noise
- Handling noise

Noise baay'inaan yoo jiraate sagalee qulqulluu dhaggeeffachuun rakkisa.

8.4. HUM MAAL Jechuudha?

Hum jechuun sagalee low-frequency elektirikii fakkaatuudha.

Yeroo baay'ee:

“MMMMMMMM...”

fakkaata.

Hum electrical grounding, power system, cable ykn equipment walitti dhufeenya irraa uumamuu danda'a.

Rakkoo grounding tokko tokko namaaf balaa elektirikii fiduu danda'a; kanaaf grounding irratti hojii sirreessaa ogeessa elektirikii ykn technician gahumsa qabuun hojjetamuu qaba.

8.5. BUZZ MAAL Jechuudha?

Buzz jechuun sagalee elektirikii:

“ZZZZZZ...”

fakkaatuudha.

Sababoota:

- Power interference
- Cable rakkoo qabu
- Connection sirrii hin taane
- Electrical equipment walitti dhiyaachuu

8.6. HISS MAAL Jechuudha?

Hiss jechuun sagalee:

“SHHHHH...”

fakkaatuudha.

Sababoota isaa keessaa:

- Gain baay’ee olkaasuu.
- Poor-quality equipment.
- Noise floor.
- Unnecessary high-frequency boost.
- Signal baay’ee xiqqaa ta’ee booda fader baay’ee olkaasuu.

2. 🎧 FAKKEENYA SAGALEE

8.7. FEEDBACK FAKKEENYA

Singer

↓

Microphone

↓

Mixer

↓

Monitor

↓

Sagalee

↓

Microphone

Yeroo monitor level baay’ee olka’u:

“IIIIII!”

fakkaatu dhaga’amuu danda’a.

8.8. HUM FAKKEENYA

Speaker irraa:

“MMMMMMMM...”

yeroo hunda dhaga’ama.

Kun hum ta’uu danda’a.

Sound Engineer cable, power, grounding fi equipment connection sakatta’a.

8.9. BUZZ FAKKEENYA

Speaker irraa:

“ZZZZZ...”

ykn:

“ZZZ-MMMM...”

dhaga’amuu danda’a.

Cable ykn electrical interference qorachuun barbaachisaa dha.

8.10. HISS FAKKEENYA

Speaker irraa:

“SHHHHH...”

yeroo signal hin jirre illee dhaga’ama.

Gain staging fi noise floor sakatta’uun barbaachisaa dha.

8.11. CABLE NOISE

XLR cable tokko sochoosuun:

“CRACK... CRACK...”

fakkaatu dhaga’amuu danda’a.

Cable ykn connector rakkoo qabaachuu danda’a.

3. □ MIXER IRRATTI AKKAMITTI HOJJETU?

8.12. FEEDBACK TO’ACHUU

Feedback yoo dhaga’ame:

1. Level saffisaan gadi buusi.
2. Microphone fi speaker placement ilaali.
3. Monitor level sakatta’i.
4. Gain sakatta’i.
5. Frequency rakkoo qabu adda baasi.
6. EQ’n frequency sana xiqqoo cut godhi.

Feedback yeroo hunda EQ qofaan furamu hin qabu.

8.13. MONITOR MIX

Monitor feedback ittisuuf:

MICROPHONE

↓

CHANNEL

↓

AUX SEND

↓

MONITOR

AUX level suuta qindeessi.

Monitor speaker microphone pickup pattern keessa akka hin seenne eegi.

8.14. NOISE TO'ACHUU

Noise yoo jiraate:

1. Source adda baasi.
2. Cable qoradhu.
3. Gain staging ilaali.
4. Power connection ilaali.
5. Unnecessary EQ boost hir'isi.
6. Faulty equipment adda baasi.

8.15. HUM/Buzz QORACHUU

Hum ykn buzz yoo jiraate:

- Cable tokko tokkoon qoradhu.
- Input tokko tokkoon adda baasi.
- Power connection sakatta'i.
- Equipment kam irraa rakkoon dhufaa akka jiru adda baasi.
- Grounding irratti hojii balaa qabu ofii keetiin hin yaalin.

8.16. GATE FI NOISE CONTROL

Mixer ykn processor tokko Noise Gate qabaachuu danda'a.

Noise Gate signal level murtaa'e gadi yoo bu'e signal sana xiqqeessuu ykn cufuu danda'a.

Fakkeenya:

Microphone

↓

Noise Gate

↓

Mixer

Garuu Noise Gate setting dogoggoraa taanaan jalqaba jecha ykn sagalee performance irraa kutuu danda'a.

4. △□ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — FEEDBACK YEROO DHAGA'AME VOLUME OLKAASUU

Kun feedback caalaatti hammeessuu danda'a.

Furmaata:

Level gadi buusi.

DOGOGGORA 2 — MICROPHONE SPEAKER FUULDURA KAAUU

Furmaata:

Microphone fi speaker kallattii sirrii irratti kaa'i.

DOGOGGORA 3 — FEEDBACK HUNDAA EQ QOFAAN FURUU

Furmaata:

Placement, gain, monitor level fi EQ hunda sakatta'i.

DOGOGGORA 4 — HUM/Buzz YEROO JIRU CABLE HUNDAA YEROO TOKKOON QORUU

Furmaata:

Cable tokko tokkoon adda baasi; source rakkoo qabu barbaadi.

DOGOGGORA 5 — GROUNDING OFII KEETIIN QORACHUU

Electrical grounding rakkoo qabu balaa elektirikii fiduu danda'a.

Furmaata:

Hojii grounding fi mains electrical safety ogeessa gahumsa qabuuf dhiisi.

DOGOGGORA 6 — NOISE HUNDAA GAIN QOFAAN FURUU

Noise sababoota hedduu qabaachuu danda'a.

Furmaata:

Source → Cable → Input → Gain → EQ → Output jechuun tartiibaan qoradhu.

DOGOGGORA 7 — NOISE GATE BAAY'EE CIMSUU

Gate baay'ee cimaa taanaan sagalee barbaachisaa illee kutuu danda'a.

Furmaata:

Threshold fi release sirriitti qindeessi.

YAADA IJOO

FEEDBACK = SPEAKER → MICROPHONE → SYSTEM LOOP

NOISE = SAGALEE HIN BARBAACHIFNE

HUM = “MMMM...”

BUZZ = “ZZZZ...”

HISS = “SHHH...”

Sound Engineer gaarii tokko rakkoo sagalee yeroo dhaga'u, mallattoo isaa qofa osoo hin ilaalle **sababa isaa adda baasee** sirreessa.

BOQONNAA 9 — DYNAMICS: COMPRESSOR, LIMITER FI NOISE GATE

1. IBSA BAL'AA

9.1. DYNAMICS PROCESSING MAAL JECUUDHA?

Dynamics Processing jechuun cimina signal audio yeroo gara garaa keessatti jijjiiramu to'achuudha.

Sagaleen tokko yeroo tokko laafaa, yeroo biraa immoo baay'ee jabaa ta'uu danda'a.

Fakkeenyaaf, nama tokko yeroo dubbatu:

“akkam...”

jedhee suuta dubbachuu danda'a; yeroo biraa immoo sagalee isaa olkaasuu danda'a.

Dynamics Processor'n garaagarummaa cimina sagalee kana to'achuuf fayyada.

Meeshaalee Dynamics Processing keessaa:

- Compressor
- Limiter
- Noise Gate

9.2. COMPRESSOR MAAL JECUUDHA?

Compressor jechuun garaagarummaa sagalee laafaa fi jabaa gidduu jiru hir'isuuf processor fayyadamnuudha.

Sagaleen baay'ee jabaan yeroo tokkoon olka'u, compressor'n isaa gadi qabu danda'a.

Bu'aan isaa:

- Level tasgabbaa'aa
- Vocal controlled
- Peaks to'atamaa
- Mix keessatti sagalee sirriitti taa'u

9.3. THRESHOLD

Threshold jechuun sadarkaa signal compressor'n hojii jalqabuudha.

Signal threshold gadi yoo ta'e:

→ Compressor hin hojjetu ykn ratio setting irratti hundaa'a.

Signal threshold ol yoo ta'e:

→ Compression jalqaba.

9.4. RATIO

Ratio'n compression cimina isaa murteessa.

Fakkeenya:

2:1

4:1

8:1

Ratio guddaan:

→ Compression cimaa.

9.5. ATTACK

Attack jechuun compressor'n signal threshold ol ga'e booda compression jalqabuuf yeroo fudhatuudha.

Attack saffisaa:

→ Peak saffisaan to'ata.

Attack dheeraa:

→ Transient tokko akka darbu eeyyama.

9.6. RELEASE

Release jechuun signal threshold gadi bu'e booda compressor'n compression dhaabuuf yeroo fudhatuudha.

Release sirrii ta'uun sagalee akka uumamaa dhaga'amu gargaara.

9.7. MAKE-UP GAIN

Compression booda signal level gadi bu'uu danda'a.

Make-up gain'n level sana deebisee olkaasuuf fayyada.

Garuu output level baay'ee olkaasuun clipping fiduu danda'a.

9.8. LIMITER

Limiter jechuun compressor fakkaata, garuu kaayyoon isaa signal peak akka daangaa murtaa'e hin caalle to'achuudha.

Limiter ratio baay'ee cimaa qabaachuu danda'a.

Fayyadama isaa:

- Peak protection
- Output protection
- Speaker/system safety keessatti gargaarsa

9.9. NOISE GATE

Noise Gate signal level threshold gadi yoo bu'e signal sana xiqqeessuu ykn cufuu danda'a.

Fakkeenya:

Microphone
↓
Noise Gate
↓
Mixer

Yeroo nama tokko hin dubbannetti background noise hir'isuuf gargaaruu danda'a.

2. FAKKEENYA SAGALEE

9.10. VOCAL FI COMPRESSOR

Nama tokko sirbaa jira.

Yeroo tokko:

“LA LA LA”

suuta sirba.

Yeroo biraa:

“LA!”

sagalee olkaasa.

Compressor'n peak ol'aanaa gadi qabuun garaagarummaa level kana hir'isa.

Bu'aan isaa:

Vocal → Controlled → Stable

9.11. BASS FI COMPRESSOR

Bass guitar keessatti note tokko tokko baay'ee jabaa, kan biraa laafaa ta'uu danda'a.

Compressor'n garaagarummaa kana hir'isuun bass akka tasgabbaa'u gargaara.

9.12. KICK FI COMPRESSOR

Kick drum tokko keessatti:

“BOOM”

fi attack isaa garaagarummaa qabaachuu danda’a.

Compressor’n transient fi body isaa to’achuuf fayyada.

9.13. LIMITER

Main output tokko keessatti peak tasaa baay’ee olka’u yoo jiraate limiter’n peak sana daangaa keessatti tursiisuuf gargaaruu danda’a.

Garuu limiter’n rakkoo mix hunda sirreessuuf miti.

9.14. NOISE GATE

Drum microphone tokko yeroo hin rukutamne:

“SHHH...”

background noise jiraachuu danda’a.

Noise Gate threshold gadi yoo kaa’ame, yeroo drum hin rukutamne noise sana hir’isuuf gargaara.

Yeroo drum rukutamu:

“BOOM!”

Gate banama.

3. MIXER IRRATTI AKKAMITTI HOJJETU?

9.15. COMPRESSOR SIGNAL CHAIN KEESSATTI

Fakkeenya:

MICROPHONE
↓
PREAMP / GAIN
↓
COMPRESSOR
↓
EQ
↓
FADER

↓
MAIN MIX

Ykn system irratti hundaa’uun:

MICROPHONE

↓
GAIN

↓
EQ

↓
COMPRESSOR

↓
FADER

↓
MAIN MIX

Tartiibni sirrii mixer ykn processor irratti hundaa’a.

9.16. COMPRESSOR QINDEESSUU

Tartiiba bu’uuraa:

1. Threshold ol irraa eegali.
2. Ratio salphaa filadhu.
3. Attack fi Release madaali.
4. Signal dhaggeeffadhu.
5. Gain Reduction ilaali.
6. Compression suuta dabali.
7. Output level sirreessi.

9.17. GAIN REDUCTION

Gain Reduction meter’n compressor’n signal meeqa gadi buusaa akka jiru agarsiisa.

Fakkeenya:

Signal peak yeroo dhufu:

→ Gain Reduction ni mul’ata.

Kun compressor hojii irra jiraachuu isaa agarsiisa.

9.18. LIMITER QINDEESSUU

Limiter yeroo output protection barbaachisutti fayyada.

Output ceiling ykn limit sirrii kaa’i.

Signal peak yeroo olka’u limiter’n daangaa sana keessatti tursiisa.

Limiter'n yeroo hunda cimaan hojii irra ooluu hin qabu.

9.19. NOISE GATE QINDEESSUU

Gate irratti:

- Threshold
- Attack
- Release
- Hold

jiraachuu danda'a.

Threshold baay'ee ol yoo ta'e:

→ Sagalee laafaa barbaachisaa kutuu danda'a.

Threshold baay'ee gadi yoo ta'e:

→ Noise hunda cufuu dhiisuu danda'a.

4. Δ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — COMPRESSOR BAAY'EE CIMSUU

Compression baay'inaan:

- Sagalee dull gochuu
- Dynamics balleessuu
- Pumping uumu
- Noise olkaasuu

danda'a.

DOGOGGORA 2 — RATIO BAAY'EE OLKAASUU

Ratio guddaan yeroo hunda gaarii miti.

Furmaata:

Kaayyoo irratti hundaa'uun ratio filadhu.

DOGOGGORA 3 — THRESHOLD DOGOGGORA

Threshold baay'ee gadi yoo ta'e compressor yeroo hunda hojii irra ooluu danda'a.

Furmaata:

Gain Reduction ilaali; gurraan dhaggeeffadhu.

DOGOGGORA 4 — MAKE-UP GAIN BAAY'EE OLKAASUU

Compression booda output olkaasuun clipping fiduu danda'a.

Furmaata:

Output meter ilaali.

DOGOGGORA 5 — LIMITER RAKKOO HUNDAAF FAYYADAMUU

Limiter mix gaarii bakka hin bu'u.

Furmaata:

Dura gain staging fi mix sirreessi.

DOGOGGORA 6 — NOISE GATE BAAY'EE CIMSUU

Gate baay'ee cimaa taanaan:

- Jecha jalqabaa kutuu
- Sustain kutuu
- Natural sound balleessuu

danda'a.

DOGOGGORA 7 — COMPRESSOR OSOO DHAGGEEFFATIN QINDEESSUU

Lakkoofsa setting qofa ilaaluun gahaa miti.

Meter + gurra lamaan fayyadami.

YAADA IJOO

COMPRESSOR

→ DYNAMICS TO'ATA

THRESHOLD

→ EESSA COMPRESSION JALQABU

RATIO

→ HAMMA COMPRESSION

ATTACK

→ YOOM COMPRESSION JALQABU

RELEASE

→ YOOM COMPRESSION DHAABU

LIMITER

→ PEAK DAANGAA KEESSATTI TO'ATA

NOISE GATE

→ SIGNAL LAFAA/NOISE TO'ATA

Sound Engineer gaarii dynamics processor fayyadamuun sagalee tasgabbaa'aa fi controlled godha, garuu processing baay'inaan fayyadamuun sagalee uumamaa hin balleessu.

BOQONNAA 10 — EFFECTS: REVERB, DELAY FI FX SEND/RETURN

1. IBSA BAL'AA

10.1. EFFECTS MAAL Jechuudha?

Effects jechuun sagalee duraan jiru irratti processing dabalataa gochuun character ykn ambience isaa jijjiiruudha.

Live Sound keessatti effects beekamoo:

- Reverb
- Delay
- Echo
- Chorus
- Flanger

Keessaa Reverb fi Delay vocal fi music irratti baay'ee fayyadamu.

Effects'n sagalee bareechuu danda'u; garuu baay'inaan fayyadamuun clarity balleessuu danda'a.

10.2. REVERB MAAL Jechuudha?

Reverb jechuun sagalee tokko room ykn hall keessa dhaga'amu fakkeessuuf reflections hedduu walitti aanan uumamuudha.

Yeroo nama tokko mana guddaa keessatti:

“Hello!”

jedhu, sagaleen isaa battalum hin dhaabbatu.

Sagaleen xiqqoo itti fufee dhaga'ama.

Kun reverb fakkeenya uumamaa dha.

10.3. REVERB GOSootA ISAA

Reverb gosa gara garaa qabaachuu danda'a:

- Room
- Hall
- Chamber
- Plate
- Spring

Room → bakka xiqqaa fakkeessa.

Hall → hall guddaa fakkeessa.

Plate → reverb artificial addaa.

Spring → texture addaa qaba.

10.4. REVERB TIME

Reverb Time ykn Decay Time jechuun reverb'n yeroo meeqa itti fufuu akka qabuudha.

Short:

→ Room xiqqaa fakkaata.

Long:

→ Hall guddaa fakkaata.

Reverb dheeraan vocal tokko mix keessatti walitti makamu danda'a.

10.5. DELAY MAAL Jechuudha?

Delay jechuun sagalee tokko yeroo muraasa booda irra deebi'ee dhaga'amuudha.

Fakkeenya:

“HELLO...”

“.....HELLO...”

Kun delay dha.

Delay'n:

- Echo effect
- Vocal effect
- Music effect
- Rhythmic effect

fayyadamuuf gargaaruu danda'a.

10.6. REVERB FI DELAY GARAAGARUMMAA

REVERB:

→ Reflections hedduu.

DELAY:

→ Repetition adda addaa.

Reverb:

“Ahhhhh...”

Delay:

“Ah... ah... ah...”

fakkaachuu danda’a.

10.7. FX SEND/RETURN

Effects external ykn internal fayyadamuuf signal channel irraa gara FX processoritti erguu dandeenya.

Sirna bu’uuraa:

CHANNEL
↓
FX SEND
↓
REVERB / DELAY
↓
FX RETURN
↓
MAIN MIX

Kun dry signal fi wet signal walitti makuuf gargaara.

2. 🎧 FAKKEENYA SAGALEE

10.8. VOCAL + REVERB

Vocal qofa:

“AMEN”

Reverb waliin:

“AMEN...en...en...”

Reverb xiqqoon vocal akka bal’atu fi bareedu godha.

Reverb baay’inaan:

→ Jechoota walitti makuu danda’a.

10.9. VOCAL + DELAY

Vocal:

“JESUS”

Delay:

“JESUS... JESUS...”

Delay yeroo muraasaaf sagalee irra deebisa.

10.10. SPEECH

Nama tokko sermon ykn speech dubbataa jira.

Speech irratti reverb baay’ee yoo dabale:

→ Jechoonni ifa hin ta’an.

Kanaaf speech irratti effect yeroo baay’ee vocal music caalaa xiqqaa ta’a.

10.11. WORSHIP VOCAL

Worship keessatti vocal irratti reverb xiqqoo ykn giddugaleessaa fayyadamuun:

- Space
- Depth
- Smoothness

dabalu danda’a.

Garuu lyrics akka ifa ta’an eeguuf barbaachisaa dha.

10.12. KEYBOARD + REVERB

Keyboard tokko keessatti effect ofii isaa qabaachuu danda’a.

Mixer irratti reverb dabaluu dura keyboard irraa effect duraan jiru ilaali.

Effects lama baay’inaan walitti makamuun sound muddy gochuu danda’a.

3. MIXER IRRATTI AKKAMITTI HOJJETU?

10.13. FX SEND QINDEESSUU

Fakkeenya:

Channel 1 = Vocal

FX Send

↓

Reverb

↓

FX Return

↓
Main Mix

10.14. REVERB QINDEESSUU

Tartiiba bu'uuraa:

1. FX processor filadhu.
2. Reverb type filadhu.
3. Channel irraa FX Send dabali.
4. Reverb Return qindeessi.
5. Main Mix keessatti dhaggeeffadhu.
6. Reverb hanga barbaachisaa ta'etti qofa dabali.

10.15. DELAY QINDEESSUU

Delay keessatti:

- Delay Time
- Feedback
- Mix/Level

jiraachuu danda'u.

Delay Time:

→ Irra deebiin yeroo meeqa booda akka dhaga'amu.

Feedback:

→ Irra deebiin meeqa akka jiraatu.

Mix/Level:

→ Effect hammam akka dhaga'amu.

10.16. DRY FI WET

Dry signal:

→ Signal original.

Wet signal:

→ Effect keessa darbe.

Fakkeenya:

Vocal dry = sagalee original.

Vocal wet = vocal + reverb/delay.

FX Return'n wet signal gara mixitti deebisa.

10.17. FX RETURN

Effect processor irraa signal deebi'u FX Return ykn Return channel irratti argamuu danda'a.

Sirni isaa:

VOCAL
↓
FX SEND
↓
REVERB
↓
RETURN
↓
MAIN MIX

10.18. EXTERNAL EFFECT PROCESSOR

Analog mixer tokko internal effects hin qabaatu taanaan external processor fayyadamuun ni danda'ama.

Mixer AUX/FX Send
↓
External Reverb/Delay
↓
Return
↓
Mixer

4. ለ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — REVERB BAAY'EE DABALUU

Reverb baay'inaan:

- Vocal ifa dhabsiisa.
- Lyrics walitti makaa.
- Mix muddy godha.

DOGOGGORA 2 — SPEECH IRRATTI REVERB BAAY'EE

Speech keessatti clarity barbaachisaa dha.

Furmaata:

Effect xiqqoo fayyadami ykn haqi.

DOGOGGORA 3 — DRY/WET HUBACHUU DHIISUU

Dry = Original.

Wet = Effect qabu.

DOGOGGORA 4 — FX SEND OLKAASUU GARUU RETURN DAGATUU

FX Send dabaluun qofti gahaa miti.

FX Return ykn effect output illee sirreessuu qaba.

DOGOGGORA 5 — DELAY FEEDBACK BAAY'EE OLKAASUU

Feedback baay'inaan:

→ Echo hedduu uuma.

→ Mix guutuu jeequu danda'a.

Furmaata:

Delay Feedback xiqqoo irraa jalqabi.

DOGOGGORA 6 — EFFECT HUNDAA VOCAL TOKKORRATTI DABALUU

Reverb + Delay + Chorus + Flanger yeroo tokkoon fayyadamuun clarity balleessuu danda'a.

Furmaata:

Effect kaayyoo qabu qofa fayyadami.

DOGOGGORA 7 — EFFECT QINDEESSUU OSOO MIX GUUTUU HIN DHAGGEEFFATIN

Solo keessatti gaarii fakkaatu effect mix keessatti baay'ee ta'uu danda'a.

Furmaata:

Effect yeroo hunda mix guutuu keessatti dhaggeeffadhu.

YAADA IJOO

REVERB

→ SPACE / ROOM FEEL

DELAY

→ REPEATED SOUND

FX SEND

→ SIGNAL GARA EFFECT ERGA

FX RETURN

→ EFFECT SIGNAL GARA MIX DEEBISA

DRY

→ ORIGINAL SIGNAL

WET

→ EFFECTED SIGNAL

Sound Engineer gaarii tokko effects sagalee dhoksuuf osoo hin taane, **sagalee tumsuuf** fayyadama.

BOQONNAA 11 — PAN, STEREO FI MIX BALANCE

1. IBSA BAL'AA

11.1. PAN MAAL Jechuudha?

PAN jechuun “Panorama” jechuudha.

Pan'n signal tokko stereo image keessatti gara:

- LEFT
- CENTER
- RIGHT

tti qooduuf fayyada.

Fakkeenya:

LEFT ← CENTER → RIGHT

Pan center yoo ta'e, signal'n Left fi Right keessatti walqixa dhaga'amu danda'a.

Pan gara Left yoo geessite, signal'n Left keessatti caalaatti dhaga'ama.

Pan gara Right yoo geessite, signal'n Right keessatti caalaatti dhaga'ama.

11.2. STEREO MAAL Jechuudha?

Stereo jechuun sagalee channel ykn image lama—Left fi Right—fayyadamuun spatial impression uumudha.

Sirna bu'uura:

LEFT + RIGHT

↓

STEREO IMAGE

Stereo mix keessatti meeshaaleen gara garaa bakka adda addaatti qoodamuu danda'u.

Garuu live sound keessatti speaker coverage, room fi audience arrangement ilaalamuu qaba.

11.3. MONO FI STEREO

MONO:

→ Sagalee tokkoon ykn signal tokkoon.

STEREO:

→ Left fi Right information adda addaa qabu.

Fakkeenya:

Vocal tokko yeroo baay'ee center irratti kaa'amuu danda'a.

Keyboard stereo output qabaachuu danda'a.

Effects tokko tokko stereo ta'uu danda'u.

11.4. CENTER IMAGE

Vocal lead yeroo baay'ee center irratti kaa'ama.

Sababni:

- Audience hundi vocal akka sirriitti dhaga'uuf.
- Main message akka ifa ta'uuf.
- Mix keessatti bakka isaa ifa gochuuf.

Fakkeenya:

VOCAL → CENTER

BASS → CENTER

KICK → CENTER

Meeshaalee biroo immoo haala mix irratti hundaa'uun pan gochuun ni danda'ama.

11.5. STEREO WIDTH

Stereo width jechuun garaagarummaa ykn bal'ina image Left fi Right gidduu jiruudha.

Width baay'ee guddaan:

→ Mix bal'aa fakkeessa.

Width baay'ee xiqqaan:

→ Mix dhiphaa fakkeessa.

Garuu width baay'inaan fayyadamuun mono compatibility rakkoo fiduu danda'a.

11.6. MONO COMPATIBILITY

Mix stereo tokko mono yoo godhame sagalee tokko tokko:

- Laafuu
- Dhabamuu
- Phase rakkoo qabaachuu

danda'u.

Kanaaf Sound Engineer mix isaa yeroo barbaachisaa ta'etti mono keessatti illee qorachuu qaba.

11.7. MIX BALANCE

Mix Balance jechuun channel hundumaa keessaa tokko kan biraa akka hin haguugne, level fi tonal relationship sirriitti qindeessuudha.

Fakkeenya:

Vocal → dominant

Keyboard → support

Guitar → support

Bass → foundation

Drum → rhythm

Balance jechuun channel hundumaa walqixa gochuu miti.

Kaayyoo music irratti hundaa'uun tokko tokko caalaa dhaga'amu danda'a.

2. 🎧 FAKKEENYA SAGALEE

11.8. VOCAL CENTER

Vocal:

LEFT ← VOCAL → RIGHT

Pan center ta'uun vocal audience keessatti bakka giddugaleessaa qaba.

11.9. GUITAR LEFT / RIGHT

Fakkeenya mix tokko keessatti:

Guitar 1 → Left

Guitar 2 → Right

Kun instruments lamaan adda baasuu danda'a.

Garuu room fi speaker coverage irratti hundaa'uun pan baay'ee extreme gochuun sirrii ta'uu dhiisuu danda'a.

11.10. KEYBOARD STEREO

Keyboard tokko Left fi Right output qabaachuu danda'a.

Keyboard:

L → Mixer Left

R → Mixer Right

Kana irraa stereo image argachuu dandeenya.

Garuu keyboard stereo output qabaachuun qofti stereo mix gaarii jechuu miti.

11.11. BASS FI KICK CENTER

Live mix keessatti bass fi kick yeroo baay'ee center irratti kaa'amuu danda'u.

Sababni:

→ Low-frequency energy gara speaker system keessatti sirriitti qooduu.

→ Mix foundation akka tasgabbaa'u.

11.12. REVERB STEREO

Reverb stereo yoo ta'e:

LEFT ← Reverb → RIGHT

fayyadamuun vocal ykn instrument tokkoof space bal'aa uumuu danda'a.

11.13. MIX BAAY'EE GUUTUU

Channel hundumaa yeroo tokkoon olkaasuun:

“WAAAA...”

fakkaatu mix uumuu danda'a.

Kun balance dhabuu agarsiisuu danda'a.

Sound Engineer tokko:

- Level
- EQ
- Pan
- Dynamics

fayyadamuun mix qulqulleessa.

3. MIXER IRRATTI AKKAMITTI HOJJETU?

11.14. PAN KNOB

Analog mixer channel tokko irratti PAN ykn BAL knob jiraachuu danda'a.

PAN CENTER:

→ Left = Right

PAN LEFT:

→ Left caala

PAN RIGHT:

→ Right caala

11.15. PAN QINDEESSUU

Tartiiba:

1. Channel filadhu.
2. Pan center irraa jalqabi.
3. Mix dhaggeeffadhu.
4. Instrument bakka addaatti barbaadde yoo ta'e gara Left ykn Rightitti sochoosi.
5. Main mix keessatti deebitee dhaggeeffadhu.

11.16. BALANCE QINDEESSUU

Dura:

- Vocal
- Kick
- Bass

akka bu'uraatti qindeessi.

Sana booda:

- Keyboard
- Guitar
- Drum
- Backing Vocal

suuta itti dabali.

Channel tokko kan biraa yoo haguugu:

→ Level ilaali.

→ EQ ilaali.

→ Pan ilaali.

11.17. STEREO KEYBOARD QINDEESSUU

Keyboard L/R yoo qabaate:

Keyboard L → CH1

Keyboard R → CH2

CH1 fi CH2 pan:

CH1 → Left

CH2 → Right

Ergasii level isaanii wal madaali.

Garuu keyboard mono yoo barbaachise, setup isaa irratti hundaa'uun output tokko fayyadamuun ni danda'ama.

11.18. MONO CHECK

Mix stereo qindaa'e booda:

→ Mono keessatti dhaggeeffadhu.

Yoo instrument tokko baay'ee laafe ykn bade:

→ Phase/correlation rakkoo jiraachuu danda'a.

Rakkoo kana adda baasuun barbaachisaa dha.

4. DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — CHANNEL HUNDAA LEFT FI RIGHTITTI EXTREME PAN GOCHUU

Kun mix artificial ykn dhiphaa taasisuu danda'a.

Furmaata:

Pan kaayyoo qabu fayyadami.

DOGOGGORA 2 — VOCAL LEAD PAN GOCHUU

Vocal lead gara extreme Left ykn Rightitti gooduun audience tokko biratti rakkoo fiduu danda'a.

Furmaata:

Vocal lead yeroo baay'ee center irratti eegi.

DOGOGGORA 3 — BASS BAAY'EE PAN GOCHUU

Low-frequency instruments yeroo baay'ee center irratti eegamuun system fi mix stability irratti faayidaa qabaachuu danda'a.

DOGOGGORA 4 — STEREO FI MONO GARAAGARUMMAA DAGATUU

Stereo → Left + Right.

Mono → Single image.

Furmaata:

Lamaan isaanii qoradhu.

DOGOGGORA 5 — MIX SOLO QOFA DHAGGEEFFATUU

Instrument tokko solo keessatti baay'ee gaarii ta'uu danda'a.

Garuu mix keessatti kan biraa haguuguu danda'a.

Furmaata:

Mix guutuu keessatti murteessi.

DOGOGGORA 6 — PAN'N BALANCE QOFA FURUU YAADUU

Pan'n channel wal irraa adda baasuu danda'a.

Garuu level fi EQ illee barbaachisaa dha.

DOGOGGORA 7 — MONO CHECK DAGATUU

Stereo mix tokko mono keessatti rakkoo qabaachuu danda'a.

Furmaata:

Yeroo barbaachisaa ta'etti mono check godhi.

YAADA IJOO

PAN = LEFT / CENTER / RIGHT QOODUU

MONO = SINGLE IMAGE

STEREO = LEFT + RIGHT IMAGE

MIX BALANCE = LEVEL + EQ + PAN + DYNAMICS WALITTI QINDEESSUU

Sound Engineer gaarii tokko pan fayyadamuun qofa osoo hin taane, **meeshaalee hundumaa keessatti bakka, level fi clarity isaanii sirriitti qindeessa.**

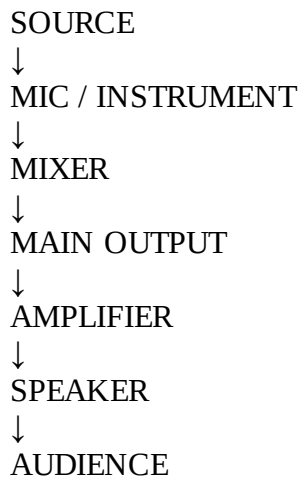
BOQONNAA 12 — AMPLIFIER, SPEAKER FI POWER MANAGEMENT

1. IBSA BAL'AA

12.1. AMPLIFIER MAAL Jechuudha?

Amplifier jechuun signal audio mixer irraa dhufu fudhatee speaker drive gochuuf humna isaa dabaluudha.

Sirni bu'uuraa:



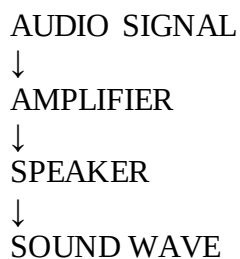
Mixer'n signal audio qindeessa.

Amplifier'n immoo signal sana speaker drive gochuuf humna barbaachisu kenna.

12.2. SPEAKER MAAL Jechuudha?

Speaker jechuun electrical audio signal gara sound wave dhaga'amutti jijjiiruudha.

Sirni isaa:



12.3. SPEAKER GOSOOTA

Live Sound keessatti speaker gosa gara garaa qaba.

- Full-range speaker
- Top speaker
- Subwoofer

- Monitor speaker
- Line-array system

Full-range speaker → frequency range bal'aa tajaajila.

Subwoofer → low frequency irratti xiyyeeffata.

Monitor → performer'f.

Top speaker → mid/high coverage irratti gahee qabaachuu danda'a.

12.4. PASSIVE FI ACTIVE SPEAKER

PASSIVE SPEAKER:

Amplifier alaa barbaada.

MIXER

↓

AMPLIFIER

↓

PASSIVE SPEAKER

ACTIVE / POWERED SPEAKER:

Amplifier keessaa qaba.

MIXER

↓

ACTIVE SPEAKER

12.5. SPEAKER IMPEDANCE

Impedance Ohm (Ω) tiin safarama.

Fakkeenya:

4 Ω

8 Ω

Amplifier fi speaker impedance wal simsiisuun barbaachisaa dha.

Amplifier manual irratti load impedance minimum isaa ilaali.

Speaker walitti parallel gochuun total impedance gadi buusuu danda'a.

Kanaaf connection speaker osoo hin godhin dura amplifier manufacturer's specification ilaali.

12.6. POWER RATING

Speaker fi amplifier irratti power rating argamuu danda'a.

Fakkeenya:

Speaker → 500 W

Amplifier → 500 W / channel

Garoo watt lakkoofsa qofa ilaaluun filannoo sirrii miti.

RMS/continuous, program fi peak ratings maal akka ta'an hubachuun barbaachisaa dha.

12.7. CLIPPING FI SPEAKER DAMAGE

Amplifier clipping yeroo baay'ee speaker irratti stress dabalu danda'a.

Amplifier baay'ee cimaan ykn signal dogoggoraan qindaa'e speaker damage fiduu danda'a.

Kanaaf:

- Gain staging
- Amplifier level
- Speaker rating
- Limiting

sirriitti to'achuu qabu.

12.8. CROSSOVER

Crossover frequency gara garaa speaker gara garaatti qooda.

Fakkeenya:

LOW

↓

SUBWOOFER

MID/HIGH

↓

TOP SPEAKER

Crossover active ykn passive ta'uu danda'a.

System tokko tokko keessatti DSP crossover fayyadamu.

2. 🎧 FAKKEENYA SAGALEE

12.9. MAIN SPEAKER

Mixer
↓
Main Output
↓
Amplifier
↓
Left Speaker
↓
Audience

Kun main PA system fakkeenya dha.

12.10. STEREO PA

Mixer L
↓
Amplifier CH1
↓
Left Speaker

Mixer R
↓
Amplifier CH2
↓
Right Speaker

12.11. SUBWOOFER + TOP

Mixer
↓
Crossover / DSP
↙ ↘
SUB TOP
↓ ↓
LOW MID/HIGH

Subwoofer low frequency tajaajila.

Top speaker mid/high coverage irratti hojjechuu danda'a.

12.12. MONITOR

Mixer AUX
↓
Monitor Output
↓
Monitor Speaker
↓
Performer

Monitor main speaker bakka hin bu'u.

12.13. AMPLIFIER CLIPPING

Mixer signal baay'ee ol

↓

Amplifier input overload

↓

Clipping

↓

Distortion

↓

Speaker stress

12.14. SPEAKER OVERLOAD

Signal ykn power speaker capability isaa caale:

→ Distortion

→ Voice coil heating

→ Mechanical stress

→ Damage

fiduu danda'a.

Kanaaf speaker rating fi amplifier setup sirriitti hubachuun barbaachisaa dha.

3. ብቅብቅ MIXER IRRATTI AKKAMITTI HOJJETU?

12.15. MAIN OUTPUT QINDEESSUU

Mixer Main L/R

↓

Amplifier Input L/R

↓

Speaker L/R

Main output level sirriitti qindeessi.

Amplifier input sensitivity fi gain/level isaa manufacturer recommendation irratti hundaa'uun qindeessi.

12.16. AMPLIFIER CONNECTION

Fakkeenya stereo system:

MIXER L



AMPLIFIER CH1



LEFT SPEAKER

MIXER R



AMPLIFIER CH2



RIGHT SPEAKER

Cable speaker sirrii fayyadami.

Speaker cable fi microphone/instrument cable bakka wal jijjiiruun sirrii miti.

12.17. SPEAKER POLARITY

Speaker polarity sirrii ta'uun barbaachisaa dha.

Speaker lama polarity wal faallaa yoo qabaatan:

→ Low frequency hir'achuu

→ Coverage rakkoo qabaachuu

→ Sound weak ta'uu

danda'a.

Kanaaf polarity fi wiring manufacturer specification irratti hundaa'uun mirkaneessi.

12.18. POWER-ON SEQUENCE

System tokko yeroo banu:

SOURCE



MIXER



PROCESSORS



AMPLIFIER / POWERED SPEAKER

Yeroo cufnu faallaa tartiiba kana fayyadamuun pop/click speaker irratti dhaga'amuuf hir'isuuf gargaara:

AMPLIFIER / POWERED SPEAKER



PROCESSORS

↓
MIXER

↓
SOURCE

12.19. SYSTEM CHECK

Live performance dura:

1. Cable ilaali.
2. Speaker connection ilaali.
3. Amplifier status ilaali.
4. Mixer output ilaali.
5. Gain staging ilaali.
6. Monitor ilaali.
7. Main speaker qoradhu.
8. Test signal ykn sound source fayyadami.
9. Sagalee dhaggeeffadhu.
10. Δ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — IMPEDANCE OSOO HIN HUBATIN SPEAKER WALITTI HIDHUU

Furmaata:

Amplifier manual fi speaker specification ilaali.

DOGOGGORA 2 — SPEAKER CABLE BAKKA MIC CABLE FAYYADAMUU

Speaker connection'f cable sirrii fayyadami.

DOGOGGORA 3 — AMPLIFIER LEVEL HUNDAA OLKAASUU

Level olkaasuun sagalee gaarii jechuu miti.

Clipping fi speaker damage fiduu danda'a.

DOGOGGORA 4 — WATT QOFAAN SYSTEM FILACHUU

Watt qofa ilaaluun gahaa miti.

Ilaali:

- Impedance
- Sensitivity
- Continuous power
- Peak capability
- Amplifier compatibility
- Room/venue size

DOGOGGORA 5 — POWER SYSTEM IRRATTI OFIIN YAALUU

Mains electricity, distribution fi grounding hojii balaa qabu.

Furmaata:

Power installation nama ogummaa qabuun haa hojjetamu.

DOGOGGORA 6 — SPEAKER FUULDURA MICROPHONE KAAUU

Feedback fiduu danda'a.

Furmaata:

Speaker fi microphone placement sirreessi.

DOGOGGORA 7 — SYSTEM OSOO TEST HIN GODHIN LIVE JALQABUU

Furmaata:

Live dura line check fi sound check godhi.

YAADA IJOO

MIXER

→ SIGNAL QINDEESSA

AMPLIFIER

→ SPEAKER DRIVE GOCHUUF HUMNA KENNA

SPEAKER

→ ELECTRICAL SIGNAL GARA SOUND JIJJIIRA

SUBWOOFER

→ LOW FREQUENCY

TOP

→ MID/HIGH

MONITOR

→ PERFORMER

Sound Engineer tokko mixer qofa beekuun gahaa miti; **mixer** → **amplifier** → **speaker** → **room** chain guutuu hubachuu qaba.

BOQONNAA 13 — CABLE, CONNECTOR FI PATCHING

1. IBSA BAL'AA

13.1. AUDIO CABLE MAAL JECUUDHA?

Audio cable jechuun electrical audio signal bakka tokko irraa gara bakka biraatti dabarsuuf fayyadamuudha.

Live Sound keessatti cable qulqulluu fi connection sirrii qabaachuun baay'ee barbaachisaa dha.

Sirni bu'uura:

SOURCE

↓

CABLE

↓

MIXER

↓

CABLE

↓

AMPLIFIER

↓

SPEAKER

13.2. XLR CABLE

XLR cable microphone fi balanced audio signal dabarsuuf baay'ee fayyada.

XLR keessatti yeroo baay'ee:

- Pin 1 → Shield/Ground
- Pin 2 → Positive/Hot
- Pin 3 → Negative/Cold

Balanced connection'n noise/interference hir'isuuf gargaara.

13.3. TRS CONNECTOR

TRS jechuun:

Tip

Ring

Sleeve

jechuudha.

TRS cable balanced audio ykn stereo connection keessatti fayyadamuu danda'a; application isaa irratti hundaa'a.

13.4. TS CONNECTOR

TS jechuun:

Tip
Sleeve

jechuudha.

TS yeroo baay'ee unbalanced instrument signal keessatti fayyada.

Fakkeenya:

- Electric guitar
- Bass guitar
- Instrument output tokko tokko

13.5. RCA CONNECTOR

RCA yeroo baay'ee consumer audio equipment keessatti argama.

Fakkeenya:

- Media player
- CD/DVD player
- Consumer audio devices

13.6. SPEAKON

Speakon connector speaker connection keessatti baay'ee fayyada.

Inni:

- Secure connection
- Speaker-level signal
- Locking mechanism

kennuu danda'a.

Speaker output amplifier irraa gara passive speakeritti geessuuf speaker connector sirrii fayyadamuun barbaachisaa dha.

13.7. BALANCED FI UNBALANCED

BALANCED:

→ Noise/interference hir'isuuf gaarii.

→ XLR fi TRS tokko tokko keessatti argama.

UNBALANCED:

→ TS/RCA fakkaatan.

→ Cable dheeraa keessatti interference irraa miidhamuu danda'a.

Application irratti hundaa'uun cable sirrii filadhu.

13.8. PATCHING MAAL JECUUDHA?

Patching jechuun source tokko gara input ykn output barbaadamutti sirnaan walqabsiisuudha.

Fakkeenya:

Microphone

↓

XLR

↓

Mixer CH1

Keyboard

↓

DI Box

↓

XLR

↓

Mixer CH2

2. 🎧 FAKKEENYA SAGALEE

13.9. VOCAL PATCH

Vocal microphone:

MIC

↓

XLR

↓

MIXER CH1

CH1:

Gain → EQ → Fader → Main Mix

13.10. KEYBOARD PATCH

Keyboard:

KEYBOARD L/R

↓

DI BOX



XLR



MIXER CH3/4

13.11. GUITAR PATCH

Electric guitar:

GUITAR



DI / AMP



MIC/LINE



MIXER

13.12. MIXER → AMPLIFIER

MAIN OUT L



XLR/TRS



AMPLIFIER CH1

MAIN OUT R



XLR/TRS



AMPLIFIER CH2

13.13. AMPLIFIER → SPEAKER

AMPLIFIER OUTPUT



SPEAKER CABLE



PASSIVE SPEAKER

Connection kana keessatti speaker cable sirrii fayyadamuun dirqama.

13.14. MONITOR PATCH

MIXER AUX 1



XLR/TRS



POWERED MONITOR

Ykn:

MIXER AUX 1

↓

AMPLIFIER

↓

PASSIVE MONITOR

13.15. FX PATCH

MIXER AUX/FX SEND

↓

EXTERNAL EFFECT

↓

RETURN

↓

MIXER

3. MIXER IRRATTI AKKAMITTI HOJJETU?

13.16. INPUT PATCH

Jalqaba channel list qopheessi.

Fakkeenya:

CH1 → Lead Vocal

CH2 → Backing Vocal

CH3 → Keyboard L

CH4 → Keyboard R

CH5 → Guitar

CH6 → Bass

CH7 → Kick

CH8 → Snare

Kun Sound Engineer'f channel saffisaan argachuuf gargaara.

13.17. LABEL GOCHUU

Channel tokkoon tokkoon isaa maqaa qabaachuu qaba.

Fakkeenya:

CH1 — VOCAL

CH2 — VOCAL 2

CH3 — KEY L

CH4 — KEY R

CH5 — GUITAR

CH6 — BASS

CH7 — KICK

CH8 — SNARE

Label sirrii rakkoo yeroo live hir'isa.

13.18. LINE CHECK

Line Check jechuun source tokkoon tokkoon isaa signal gara mixeritti dhufaa jiraachuu qorachuudha.

Tartiiba:

MIC CH1 → TEST

MIC CH2 → TEST

KEYBOARD → TEST

GUITAR → TEST

BASS → TEST

DRUM → TEST

Signal dhufuu isaa mirkaneessi.

13.19. CABLE TEST

Signal yoo hin dhufne:

1. Source ilaali.
2. Cable ilaali.
3. Connector ilaali.
4. Mixer input ilaali.
5. Gain ilaali.
6. Mute ilaali.
7. Fader ilaali.
8. Output route ilaali.

Rakkoo tokko tokkoon adda baasi.

13.20. CABLE MANAGEMENT

Cable live stage irratti:

- Walitti hin makamin.
- Karaa namni irra deemutti hin diriirin.
- Power cable fi audio cable haala sirriin qindeessi.
- Cable coil sirriitti godhi.
- Connector hin miidhin.

Cable management gaariin safety fi reliability fooyyeessa.

4. △□ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — MIC CABLE FI SPEAKER CABLE WALITTI MAKUU

Speaker-level output cable sirrii barbaada.

Furmaata:

Cable irratti maqaa fi type isaa hubadhu.

DOGOGGORA 2 — XLR PIN DOGOGGORAA HOJJETUU

Pin assignment sirrii hin taane noise ykn phase/polarity rakkoo fiduu danda'a.

Furmaata:

Standard wiring fi manufacturer specification hordofi.

DOGOGGORA 3 — CABLE CABINET KEESSA WALITTI MAKUU

Cable walitti makamuun troubleshooting rakkisa.

Furmaata:

Cable label godhi.

DOGOGGORA 4 — CABLE DHOORAA FAYYADAMUU

Cable dhooraan:

- Crackle
- Signal loss
- Hum
- Intermittent signal

fiduu danda'a.

Furmaata:

Cable qoradhu ykn bakka buusi.

DOGOGGORA 5 — LINE CHECK DHIISUU

Live performance jalqabee booda cable rakkoo argachuun rakkoo guddaa dha.

Furmaata:

Soundcheck dura line check godhi.

DOGOGGORA 6 — CABLE POWER FI AUDIO WALITTI MAKUU

Electrical interference ykn safety rakkoo fiduu danda'a.

Furmaata:

Cable routing sirriitti qindeessi.

DOGOGGORA 7 — CONNECTOR DIRQISIISUU

Connector tokko yoo hin seene humnaan hin galchin.

Furmaata:

Type fi direction isaa mirkaneessi.

DOGOGGORA 8 — PATCH SHEET DHIISUU

Setup guddaa keessatti channel maal akka ta'e dagachuun salphaa dha.

Furmaata:

Patch list ykn channel list qopheessi.

YAADA IJOO

XLR → MICROPHONE / BALANCED AUDIO

TRS → BALANCED AUDIO / STEREO APPLICATION TOKKO TOKKO

TS → UNBALANCED INSTRUMENT

RCA → CONSUMER AUDIO

SPEAKON → SPEAKER CONNECTION

PATCHING → SIGNAL ROUTING

LINE CHECK → SIGNAL HUNDAA QORACHUU

Sound Engineer gaarii tokko **cable sirrii + connector sirrii + patching sirrii + labeling sirrii** fayyadamuun hojii isaa saffisaa fi amanamaa taasisa.

BOQONNAA 14 — GAIN STAGING FI SIGNAL FLOW

1. IBSA BAL'AA

14.1. GAIN STAGING MAAL JECUUUDHA?

Gain Staging jechuun signal audio system guutuu keessa darbu keessatti sadarkaa isaa bakka tokkoon tokkoon isaa sirriitti qindeessuudha.

Kaayyoon isaa:

- Noise hir'isuu
- Distortion ittisuu
- Clarity eeguu
- Headroom qabaachuu
- Mixer, amplifier fi speaker sirriitti hojjechiisuu

Signal chain fakkeenya:

MICROPHONE

↓

PREAMP / GAIN

↓

EQ

↓

FADER

↓

MAIN BUS

↓

OUTPUT

↓

AMPLIFIER

↓

SPEAKER

14.2. GAIN FI FADER GARAAGARUMMAA

GAIN:

Signal input level qindeessa.

FADER:

Signal level mix keessatti qindeessa.

Kana jechuun:

GAIN = INPUT CONTROL

FADER = MIX CONTROL

Gain baay'ee olkaasuun fader gadi buusuu qofa sirna gaarii miti.

14.3. HEADROOM MAAL Jechuudha?

Headroom jechuun signal normal operating level fi clipping point gidduu bakka nageenyaa jiruudha.

Signal:

NORMAL LEVEL

↓

HEADROOM

↓

CLIPPING

Headroom gahaan peak tasaa dhufuuf bakka kenna.

14.4. CLIPPING MAAL Jechuudha?

Clipping jechuun signal system capacity isaa yeroo caalu waveform isaa cituu ykn distortion uumamuudha.

Sagaleen:

“Clean”

irraa gara:

“CRUNCH / DISTORTED”

tti jijjiiramuu danda'a.

Clipping analog system keessatti signal level baay'ee olkaasuu irraa uumamuu danda'a.

14.5. NOISE FLOOR

Noise floor jechuun noise system keessatti yeroo signal barbaachisaan hin jirretti uumamuudha.

Gain sirrii hin taane:

→ Noise olkaasuu danda'a.

Signal source baay'ee xiqqaa ta'ee gain stages booda baay'ee olkaasuun noise dabalu danda'a.

14.6. SIGNAL-TO-NOISE RATIO

Signal-to-Noise Ratio jechuun signal barbaachisaa fi noise gidduu garaagarummaa agarsiisa.

Signal qulqulluu fi noise gadi aanaa:

→ SNR gaarii.

Noise signal waliin wal bira qabamee baay'ee yoo ta'e:

→ SNR gadi aanaa.

14.7. UNITY GAIN

Fader tokko irratti "0" ykn unity position jiraachuu danda'a.

Unity jechuun yeroo baay'ee signal level nominal irratti attenuation ykn boost guddaa malee dabarsuuf reference point ta'a.

Garuu mixer hundi design tokko miti.

2. 🎧 FAKKEENYA SAGALEE

14.8. GAIN DOGOGGORA

Microphone

↓

GAIN BAAY'EE OL

↓

DISTORTION

↓

MIXER

Sagaleen microphone irraa dhufu jalqabuma irratti clipping yoo godhe, fader gadi buusuun distortion sana hin balleessu.

Sababni:

Rakkoon input stage irratti uumame.

14.9. GAIN XIQQAA

Microphone

↓

GAIN BAAY'EE XIQQAA

↓

SIGNAL LAFAA

↓

FADER BAAY'EE OL

↓

NOISE

Kun noise floor akka caalaatti dhaga'amu gochuu danda'a.

14.10. GAIN SIRRII

MIC
↓
GAIN SIRRII
↓
CLEAN SIGNAL
↓
FADER
↓
MAIN MIX

Kun signal qulqulluu fi headroom gaarii argamsiisa.

14.11. VOCAL FAKKEENYA

Singer:

“LA LA LA”

Yeroo sagalee isaa olkaasu:

“LAAAA!”

Gain sirriitti qindaa’e yoo ta’e peak kun headroom keessatti hafa.

Gain baay’ee ol yoo ta’e:

“LAAAA!” → CLIP

14.12. KEYBOARD FAKKEENYA

Keyboard output baay’ee jabaa ta’uu danda’a.

Keyboard
↓
Mixer Input
↓
GAIN SIRRII
↓
CLEAN SIGNAL

Gain baay’ee olkaasuun input overload fiduu danda’a.

14.13. BASS FI DRUM

Bass fi drum peak cimaa qabaachuu danda’u.

Kanaaf meter ilaaluun qofti osoo hin taane, peak yeroo performance dhufu dhaggeeffachuu fi headroom eegu barbaachisaa dha.

3. ብቅ MIXER IRRATTI AKKAMITTI HOJJETU?

14.14. GAIN SET GOCHUU

Tartiiba bu'uuraa:

1. Channel fader bakka hojii isaa kaa'i.
2. Main output level sirrii qopheessi.
3. Performer'n level performance isaa dhugaa fakkaatuun haa taphatu.
4. PFL/Solo meter yoo jiraate ilaali.
5. Gain suuta olkaasi.
6. Peak yeroo performance dhufu ilaali.
7. Clipping irraa fagaadhu.
8. Headroom gahaa dhiisi.

14.15. PFL / SOLO

Mixer tokko PFL ykn Solo qabaachuu danda'a.

PFL = Pre-Fader Listen.

Kun input signal fader dura dhaggeeffachuu fi meter irratti ilaaluuf gargaaruu danda'a.

Sound Engineer gain setting keessatti PFL/Solo fayyadamuun signal input sirriitti hubachuu danda'a.

14.16. FADER QINDEESSUU

Gain sirriitti erga qindaa'ee booda:

→ Fader'n mix balance qindeessa.

Fader olkaasuun signal input rakkoo qabu sirreessuuf mala jalqabaa miti.

14.17. MAIN OUTPUT

Channel hundi:

CHANNEL FADERS

↓

MAIN BUS

↓

MAIN FADER

↓

OUTPUT

Main output meter ilaali.

Output clip yoo godhe:

- Channel levels
- Main bus
- Processing
- Output gain

sakatta'i.

14.18. AMPLIFIER INPUT

Mixer output qulqulluu ta'ee gara amplifieritti seena.

MIXER

↓

CLEAN OUTPUT

↓

AMPLIFIER INPUT

↓

SPEAKER

Amplifier input level fi mixer output wal simsiisuu qaba.

Amplifier input overload gochuun distortion fiduu danda'a.

14.19. SIGNAL FLOW QORACHUU

Sagaleen yoo hin dhufne, tartiiba kana hordofi:

SOURCE

↓

CABLE

↓

INPUT

↓

GAIN

↓

CHANNEL

↓

BUS

↓

MAIN

↓

OUTPUT

↓

AMPLIFIER

↓

SPEAKER

Bakka signal itti badu adda baasi.

4. Δ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — GAIN FI FADER WAL QIXA SE'UU

Gain input qindeessa.

Fader mix qindeessa.

Lamaan hojii adda qabu.

DOGOGGORA 2 — CLIPPING ERGA JIRU BOODA FADER GADI BUUSUU

Input clipping yoo jiraate fader gadi buusuun rakkoo input sana hin sirreessu.

Furmaata:

Gain input sirreessi.

DOGOGGORA 3 — GAIN BAAY'EE XIQQAA + FADER BAAY'EE OL

Signal weak ta'ee booda boost baay'inaan gochuun noise olkaasuu danda'a.

DOGOGGORA 4 — METER QOFA ILAALUU

Meter barbaachisaa dha.

Garuu gurraan dhaggeeffachuunis murteessaa dha.

DOGOGGORA 5 — HEADROOM DHIISUU DHIISUU

Signal yeroo hunda 0 dB ykn clipping bira tursiisuun nageenya hin qabu.

Peak'f bakka dhiisi.

DOGOGGORA 6 — MAIN OUTPUT CLIP

Channel tokko qofa osoo hin taane main bus/output illee clip gochuu danda'a.

Furmaata:

Signal chain guutuu sakatta'i.

DOGOGGORA 7 — SIGNAL FLOW HUBACHUU DHIISUU

Sound Engineer tokko rakkoo yoo uumame “maalif sagaleen hin jiru?” qofa jechuu osoo hin taane:

“Signal eessa irraa dhufee eessa irratti bade?”

jedhee barbaaduu qaba.

YAADA IJOO

GAIN = INPUT LEVEL

FADER = MIX LEVEL

HEADROOM = BAKKA PEAK'F HAFU

CLIPPING = SIGNAL CAPACITY CAALUU

NOISE FLOOR = SYSTEM NOISE

PFL = INPUT SIGNAL DHAGGEEFFACHUU / ILAALUU

SIGNAL FLOW = SOURCE → MIXER → OUTPUT → AMPLIFIER → SPEAKER

Gain staging sirrii jechuun **signal bakka hundatti qulqulluu, controlled fi headroom gahaa qabu** gochuudha.

BOQONNAA 15 — EQ (EQUALIZATION) FI FREQUENCY CONTROL

1. IBSA BAL'AA

15.1. EQ MAAL Jechuudha?

EQ jechuun **Equalization** jechuudha.

EQ'n frequency adda addaa signal audio keessa jiran olkaasuu ykn gadi buusuuf fayyada.

Sagalee namaa fi meeshaalee muuziqaa hundi frequency gara garaa qabu.

Fakkeenya:

LOW → BASS

MID → BODY / CLARITY

HIGH → TREBLE / DETAIL

Sound Engineer'n EQ fayyadamuun:

- Sagalee ifa gochuu
- Muddy sound hir'isuu
- Harshness hir'isuu
- Feedback frequency tokko to'achuu
- Instruments wal irraa adda baasuu

danda'a.

15.2. FREQUENCY MAAL Jechuudha?

Frequency jechuun yeroo tokko keessatti waveform'n meeqa akka irra deddeebi'uudha.

Unit isaa:

Hertz (Hz)

Fakkeenya:

50 Hz → Low frequency

500 Hz → Mid frequency

5,000 Hz → High frequency

20,000 Hz → High frequency baay'ee ol'aanaa

Namni dhagahuu danda'u range isaa nama irraa namatti garaagarummaa qaba; akka waliigalaatti hearing namaa 20 Hz hanga 20 kHz naannoo isaa keessatti ibsama.

15.3. LOW FREQUENCY

Low frequency keessatti:

- Kick
- Bass
- Deep sound
- Rumble

argamuu danda'u.

Yeroo baay'ee:

20–100 Hz

naannoo keessatti energy gadi fagoo argama.

Low baay'inaan:

→ Boomy

→ Muddy

ta'uu danda'a.

15.4. MID FREQUENCY

Mid frequency keessatti:

- Vocal body
- Guitar
- Keyboard
- Snare
- Speech information

baay'inaan argamu.

Mid baay'inaan:

→ Boxy

→ Muddy

→ Nasal

ta'uu danda'a.

15.5. HIGH FREQUENCY

High frequency keessatti:

- Vocal detail
- Cymbal
- Air
- Brightness
- Attack

argamuu danda'a.

High baay'inaan:

→ Harsh

→ Sharp

→ Fatiguing

ta'uu danda'a.

15.6. BOOST FI CUT

BOOST:

→ Frequency olkaasuu.

CUT:

→ Frequency gadi buusuu.

Sound Engineer gaarii yeroo baay'ee frequency hin barbaachifne **cut** gochuun jalqaba.

Sababni:

Frequency barbaachisaa qofa olkaasuun mix qulqulleessuu danda'a.

15.7. PARAMETRIC EQ

Parametric EQ keessatti yeroo baay'ee:

- Frequency
- Gain
- Q

to'achuu dandeenya.

Frequency → bakka EQ hojjetu.

Gain → hamma boost/cut.

Q → bal'ina ykn dhiphachuu bandwidth.

15.8. HIGH-PASS FILTER (HPF)

HPF jechuun frequency gadi aanaa murteessitoota gadi buusu ykn haquudha.

Fakkeenya:

Vocal keessatti rumble fi handling noise hir'isuuf fayyada.

Garuu HPF baay'ee olkaasuun vocal body hir'isuu danda'a.

2. FAKKEENYA SAGALEE

15.9. VOCAL MUDDY

Vocal:

“WAAAA...”

“UUMMM...”

fakkaatee dhaga'ama.

Low-mid baay'achuu danda'a.

EQ'n frequency rakkoo qabu xiqqoo cut gochuun vocal ifa gochuu danda'a.

15.10. VOCAL DULL

Vocal detail dhabee:

“mm... mm...”

fakkaata.

High-frequency detail xiqqoo ta'uu danda'a.

Garuu boost gochuu dura microphone, placement fi gain sakatta'i.

15.11. VOCAL HARSH

Vocal:

“SSSS!”

“SHHH!”

“AAAA!”

fakkaatee gurra namaa qara.

High-mid ykn high frequency baay'achuu danda'a.

Frequency rakkoo qabu adda baasuun xiqqoo cut godhuun ni danda'ama.

15.12. KICK DRUM

Kick keessatti:

LOW → Body

MID → Boxiness

HIGH/MID-HIGH → Attack

argamuu danda'a.

Kick tokko:

“BOOM”

fi

“CLICK”

lamaan qabaachuu danda'a.

15.13. BASS

Bass keessatti low frequency bu'uura isaa ti.

Garuu low-mid baay'inaan:

→ Muddy

→ Cloudy

gochuu danda'a.

Bass fi kick frequency walitti bu'an yoo ta'e EQ fi arrangement fayyadamuun separation fooyyeessuu danda'a.

15.14. GUITAR

Guitar keessatti:

Low → Body

Mid → Character

High → Detail

argamu danda'a.

Guitar low baay'inaan mix keessa bass fi kick haguuguu danda'a.

15.15. KEYBOARD

Keyboard stereo ykn mono ta'uu danda'a.

Low:

→ Body

Mid:

→ Chords / notes

High:

→ Detail / brightness

Keyboard baay'ee guutuu taanaan instruments biroo haguuguu danda'a.

3. MIXER IRRATTI AKKAMITTI HOJJETU?

15.16. CHANNEL EQ

Analog mixer tokko irratti EQ channel:

HIGH

MID

LOW

ta'uu danda'a.

Mixer tokko tokko:

HIGH

MID

LOW

qofa qaba.

Kan biraa:

HIGH

MID

LOW-MID

HIGH-MID

qabaachuu danda'a.

15.17. HPF QINDEESSUU

Vocal irratti HPF yoo jiraate:

1. HPF banuu.
2. Frequency isaa suuta olkaasuu.
3. Vocal body dhaggeeffachuu.
4. Rumble hir'achuu ilaali.
5. Vocal natural ta'ee akka hafu mirkaneessi.

Lakkoofsi HPF sirrii microphone, singer, room fi application irratti hundaa'a.

15.18. MUDDY SOUND SIRREESSUU

Yoo mix muddy ta'e:

1. Channel adda baasi.
2. Low-mid sakatta'i.
3. Frequency rakkoo qabu barbaadi.
4. Xiqqoo cut godhi.
5. Mix guutuu keessatti dhaggeeffadhu.

Channel hundarra frequency tokko tokko cut gochuun yeroo hunda furmaata miti.

15.19. FEEDBACK EQ

Feedback frequency tokko irra deddeebi'ee yoo dhaga'ame:

1. Level nageenya qabu eegi.
2. Frequency rakkoo qabu adda baasi.
3. EQ ykn graphic EQ irratti frequency sana xiqqoo cut godhi.
4. Monitor placement sakatta'i.
5. Gain fi AUX level deebi'ee ilaali.

Feedback sirreessuuf EQ qofa irratti hirkachuun sirrii miti.

15.20. GRAPHIC EQ

Graphic EQ keessatti frequency bands hedduu qaba.

Fakkeenya:

63 Hz
125 Hz
250 Hz
500 Hz
1 kHz
2 kHz
4 kHz

8 kHz
16 kHz

Band tokkoon tokkoon isaa boost ykn cut gochuu dandeessa.

Live sound keessatti monitor fi main PA tuning irratti fayyada.

15.21. PARAMETRIC EQ FI GRAPHIC EQ

PARAMETRIC EQ:

→ Frequency filachuu

→ Gain jijjiiruu

→ Q jijjiiruu

GRAPHIC EQ:

→ Frequency bands murtaa'an irratti boost/cut

4. △□ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — EQ HUNDAA BOOST GOCHUU

High + Mid + Low hundumaa olkaasuun mix guutuu fi muddy gochuu danda'a.

Furmaata:

EQ'n rakkoo sirreessuuf fayyadami.

DOGOGGORA 2 — FREQUENCY HIN DHAGGEEFFATIN LAKKOOFSA QOFAAN HOJJETUU

Lakkoofsi EQ qofa gahaa miti.

Gurra kee fayyadami.

DOGOGGORA 3 — HPF BAAY'EE OLKAASUU

HPF baay'ee ol yoo ta'e low-end barbaachisaa kutuu danda'a.

DOGOGGORA 4 — VOCAL IFSA GOCHUUF HIGH BAAY'EE BOOST GOCHUU

Kun harshness fi ear fatigue fiduu danda'a.

Furmaata:

Frequency rakkoo qabu adda baasi.

DOGOGGORA 5 — EQ CHANNEL HUNDAA WAL FAKKAATTI QINDEESSUU

Microphone fi source hundi wal fakkaatu miti.

Furmaata:

Channel tokkoon tokkoon isaa dhaggeeffadhu.

DOGOGGORA 6 — SOLO QOFAAN EQ GOCHUU

Solo keessatti gaarii fakkaatu mix keessatti rakkoo qabaachuu danda'a.

Furmaata:

Solo → qorannoo

Full Mix → murtii dhumaa

DOGOGGORA 7 — EQ'N RAKKOO PLACEMENT FURUU YAALUU

Microphone bakka dogoggoraa yoo jiraate EQ baay'inaan fayyadamuun furmaata guutuu miti.

Furmaata:

Dura source fi placement sirreessi.

YAADA IJOO

EQ = FREQUENCY CONTROL

LOW = BODY / BASS

MID = BODY / CHARACTER / CLARITY

HIGH = DETAIL / BRIGHTNESS

BOOST = OLKAASUU

CUT = GADI BUUSUU

HPF = LOW FREQUENCY HIN BARBAACHISNE HIR'ISUU

Sound Engineer gaarii tokko EQ fayyadamuun sagalee **caalaatti ifa, balanced fi qulqulluu** godha; EQ baay'inaan fayyadamuun garuu sagalee uumamaa balleessuu danda'a.

BOQONNAA 16 — SOUNDCHECK FI LINE CHECK

1. IBSA BAL'AA

16.1. LINE CHECK MAAL Jechuudha?

Line Check jechuun source sagalee tokkoon tokkoon isaa irraa signal gara mixeritti sirriitti dhufaa jiraachuu mirkaneessuudha.

Line Check keessatti:

Microphone
Keyboard
Guitar
Bass
Drum
Playback

fi source biroo tokkoon tokkoon isaanii qoramu.

Sirni isaa:

SOURCE
↓
CABLE
↓
MIXER INPUT
↓
SIGNAL
↓
METER

Kaayyoon isaa rakkoo cable, connector, input ykn routing live performance dura adda baasuu dha.

16.2. SOUNDCHECK MAAL Jechuudha?

Soundcheck jechuun system guutuu fi performers waliin sagalee live performance'f qopheessuudha.

Soundcheck keessatti:

- Gain
- EQ
- Dynamics
- Monitor
- Main PA
- Effects
- Balance

qoratamu.

16.3. LINE CHECK FI SOUNDCHECK GARAAGARUMMAA

LINE CHECK:

→ Signal jiraachuu isaa qofa mirkaneessa.

SOUNDCHECK:

→ Sagalee source sanaa mix keessatti sirreessa.

Fakkeenya:

Line Check:

“CH1 signal jira?”

Soundcheck:

“CH1 vocal level, EQ fi monitor sirrii dha?”

16.4. SOUNDCHECK DURA

Soundcheck dura:

- Stage setup xumuruu.
- Microphone kaa’uu.
- Cable hunda connect gochuu.
- Mixer setup qopheessuu.
- Main speaker qopheessuu.
- Monitor qopheessuu.
- Power system sirrii ta’uu mirkaneessuu.

Power fi electrical safety hojii isaa ogeessa gahumsa qabuun mirkanaa’uu qaba.

16.5. CHANNEL LIST

Sound Engineer channel list qabaachuu qaba.

Fakkeenya:

CH 1 — Lead Vocal

CH 2 — Backing Vocal

CH 3 — Keyboard L

CH 4 — Keyboard R

CH 5 — Guitar

CH 6 — Bass

CH 7 — Kick

CH 8 — Snare

CH 9 — Overhead L

CH 10 — Overhead R

Channel list'n hojii saffisaa fi sirrii taasisa.

16.6. STAGE PLOT

Stage Plot jechuun meeshaaleen stage irratti eessa akka jiran agarsiisuudha.

Fakkeenya:

VOCAL → CENTER

KEYBOARD → LEFT

GUITAR → RIGHT

DRUM → REAR CENTER

MONITOR → PERFORMER FUULDURA

Stage plot qabaachuun setup saffisa.

2. 🎧 FAKKEENYA SAGALEE

16.7. VOCAL LINE CHECK

Sound Engineer:

“Vocal 1, check please.”

Singer:

“Hello... one, two...”

Meter:

→ Signal jira.

Kana booda:

Gain



EQ



Fader



Main

16.8. KEYBOARD LINE CHECK

Keyboardist:

“Check...”

Sound Engineer:

→ CH3 L signal

→ CH4 R signal

sakatta’a.

Left fi Right lamaan isaanii sirrii ta’uu qabu.

16.9. GUITAR LINE CHECK

Guitarist:

“Check...”

Sound Engineer:

→ Signal

→ Noise

→ Gain

qorata.

16.10. DRUM LINE CHECK

Drummer ykn technician tokko:

Kick → “Boom”

Snare → “Crack”

Tom → “Tom”

Cymbal → “Chh”

jechuun source tokkoon tokkoon isaa test godha.

16.11. MONITOR CHECK

Performer:

“Monitor koo keessatti vocal xiqqoo dabali.”

Sound Engineer:

AUX

↓

Vocal Send

↓

Monitor

qindeessa.

16.12. MAIN PA CHECK

Sound Engineer main PA dhaggeeffata.

Left:

→ Signal

Right:

→ Signal

L/R balance, coverage fi unwanted noise sakatta’a.

16.13. FEEDBACK CHECK

Monitor level suuta olkaasuun system feedback itti dhiyaatu adda baasuu danda’a.

Garuu test kun:

- Namaaf balaa hin fidne.
- Speaker damage hin fidne.
- Level nageenya qabu keessatti

raawwatamuu qaba.

Feedback uumamu yoo jalqabe level gadi buusi.

3. MIXER IRRATTI AKKAMITTI HOJJETU?

16.14. LINE CHECK TARTIIBA

1. Channel 1 irraa jalqabi.
2. Source test godhi.
3. Meter ilaali.
4. Gain sirreessi.
5. Channel mute status ilaali.
6. Fader hojii isaa ilaali.
7. Main route ilaali.
8. Channel itti aanu deemi.

16.15. GAIN SETTING

Performer'n akka performance dhugaa fakkaatutti taphatu/dubbatu haa godhu.

Gain suuta olkaasi.

Peak yeroo dhufu ilaali.

Clipping irraa fagaadhu.

Headroom dhiisi.

16.16. EQ CHECK

Gain erga sirrii ta'ee:

→ HPF yoo barbaachise qindeessi.

→ Low/mid/high balance ilaali.

→ Frequency rakkoo qabu xiqqoo cut godhi.

EQ baay'inaan boost hin godhin.

16.17. MONITOR MIX

AUX tokko tokkoon:

AUX 1 → Vocal Monitor

AUX 2 → Keyboard Monitor

AUX 3 → Band Monitor

fakkeenya ta'uu danda'a.

Performer tokkoon tokkoon isaa maal akka barbaadu gaafadhu.

16.18. FX CHECK

Vocal:

→ Reverb

→ Delay

qoradhu.

Effect baay'ee ta'uu isaa mix guutuu keessatti dhaggeeffadhu.

16.19. MAIN MIX CHECK

Main Mix keessatti:

- Vocal
- Instrument
- Bass
- Drum
- Effects

walitti dhufeenya isaanii ilaali.

Kaayyoon:

→ Vocal ifa

→ Music balanced

→ Low-end controlled

→ High-end hin qarin

→ Feedback hin jirre

16.20. FINAL CHECK

Performance dura:

✓ Channel hundi signal qaba.

✓ Gain sirrii.

✓ EQ sirrii.

✓ Monitor sirrii.

✓ Main PA sirrii.

✓ FX sirrii.

✓ Cable sirrii.

✓ Backup cable jira.

✓ Power system nageenya qaba.

✓ Channel list qophaa'e.

4. ⚠ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — LINE CHECK DHIISUU

Live performance jalqabee cable rakkoo argachuun rakkoo guddaa dha.

Furmaata:

Channel hundumaa dura qoradhu.

DOGOGGORA 2 — SOUNDCHECK YEROO PERFORMER HIN JIRRETTI QOFATTI HOJJETUU

Performer dhugaan sagalee isaa adda ta'e qaba.

Furmaata:

Yeroo danda'ametti performer mataa isaa waliin soundcheck godhi.

DOGOGGORA 3 — SOUNDCHECK IRRATTI EFFECT BAAY'EE DABALUU

Soundcheck keessatti effect baay'inaan dabaluun mix dhoksuu danda'a.

Furmaata:

Dura dry signal sirreessi; booda effect dabali.

DOGOGGORA 4 — MONITOR QORACHUU DHIISUU

Main PA gaarii ta'us monitor rakkoo qabaachuu danda'a.

Furmaata:

Monitor tokkoon tokkoon isaa qoradhu.

DOGOGGORA 5 — CHANNEL LIST QABAACHUU DHIISUU

Channels hedduu keessatti rakkoo saffisaan adda baasuun rakkisa.

Furmaata:

Channel list fi label fayyadami.

DOGOGGORA 6 — FEEDBACK TEST BAAY'EE OLKAASUU

System ykn nama irratti balaa fiduu danda'a.

Furmaata:

Level nageenya qabuun hojjedhu.

DOGOGGORA 7 — SOUND ENGINEER QOFA DHAGGEEFFATUU

Performer'n monitor isaa maal akka barbaadu caalaatti beeka.

Furmaata:

Performer waliin communicate godhi.

YAADA IJOO

LINE CHECK

→ SIGNAL JIRAACHUU MIRKANEESSUU

SOUNDCHECK

→ SAGALEE FI MIX SIRREESSUU

CHANNEL LIST

→ CHANNEL HUBACHUUF

STAGE PLOT

→ STAGE SETUP HUBACHUUF

MONITOR CHECK

→ PERFORMER'F

MAIN PA CHECK

→ AUDIENCE'F

Soundcheck gaariin performance gaarii jalqabuuf bu'uura. Sound Engineer tokko **live jalqabuun dura rakkoo adda baasuu** qaba; live jalqabee booda rakkoo barbaaduu caalaa kun hojii professional dha.

BOQONNAA 17 — STAGE SETUP FI MICROPHONE PLACEMENT

1. IBSA BAL'AA

17.1. STAGE SETUP MAAL JECUUUDHA?

Stage Setup jechuun meeshaalee sagalee, microphone, instrument, monitor fi speaker bakka sirrii irratti qindeessuudha.

Stage setup gaarii:

- Sagalee qulqulluu
- Feedback xiqqaa
- Hojii saffisaa
- Safety fooyya'aa
- Performers'f mijataa

fida.

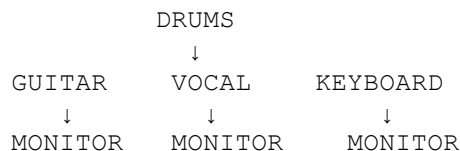
Sirna bu'uuraa:

STAGE
↓
MICROPHONES + INSTRUMENTS
↓
CABLES
↓
MIXER
↓
MAIN PA

17.2. STAGE PLOT

Stage Plot jechuun fakkii ykn karoora stage irratti meeshaaleen eessa akka kaa'aman agarsiisuudha.

Fakkeenya:



Stage plot'n setup yeroo hunda wal fakkaataa gochuuf gargaara.

17.3. MICROPHONE PLACEMENT

Microphone placement jechuun microphone source sagalee irraa eessa fi akkam fageenya irratti akka kaa'amuudha.

Placement sirrii:

→ Signal qulqulluu

→ Room noise xiqqaa

→ Feedback xiqqaa

→ EQ hojii xiqqaa

fiduu danda'a.

17.4. MICROPHONE FI SOURCE FAGEENYA

Microphone source irraa baay'ee fagoo yoo ta'e:

- Room sound dabala.
- Noise dabala.
- Direct signal hir'isa.

Baay'ee dhihoo yoo ta'e:

- Proximity effect
- Plosive
- Uneven level

uumamuu danda'a.

Kanaaf fageenyi madaalawaa barbaachisaa dha.

17.5. SPEAKER PLACEMENT

Main speaker audience gara:

→ Fuuldura

→ Gara audience

itti qajeeluu qaba.

Microphone pickup area keessa speaker direct akka hin seene eegi.

Speaker placement sirrii feedback fi coverage irratti dhiibbaa qaba.

17.6. MONITOR PLACEMENT

Monitor speaker yeroo baay'ee performer fuuldura, microphone pickup pattern isaa irratti hundaa'uun kaa'ama.

Kaayyoon:

Performer'n sagalee barbaachisaa dhaga'uu.

Monitor speaker microphone duuba ykn kallattii pickup isaa keessatti kaa'amuun yeroo baay'ee feedback hir'isuuf gargaara; garuu microphone pattern gara garaa qaba, kanaaf placement microphone-specific ta'uu qaba.

17.7. CABLE ROUTING

Cable:

- Karaa namni irra deemu irraa fagaachuu.
- Tape ykn cable ramp fayyadamuun yeroo barbaachisaa ta'etti secure gochuu.
- Power fi audio cable haala nageenya qabuun qindeessuu.
- Connector irratti tension hin uumin.

Cable routing gaariin stage safety fooyyeessa.

2. 🎧 FAKKEENYA SAGALEE

17.8. VOCAL STAGE

Singer:



Monitor singer fuuldura jira.

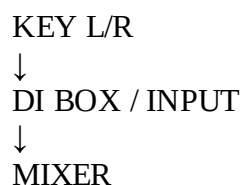
Microphone fi monitor placement feedback hir'isuuf qindaa'a.

17.9. KEYBOARD STAGE

Keyboardist:



Keyboard output:



17.10. GUITAR STAGE

Guitar:

GUITAR

↓

AMP

↓

MIC

↓

MIXER

Amp fi microphone bakka wal simutti kaa'amu.

Amp audience ykn performer barbaadetti qajeeluu qaba.

17.11. DRUM STAGE

Drum setup fakkeenya:

KICK

SNARE

HI-HAT

TOM

OVERHEAD

Microphone tokkoon tokkoon isaa source isaa irratti xiyyeeffata.

Cable hunda drum area keessa walitti makamu irraa eegi.

17.12. FULL BAND

Fakkeenya:

DRUM

↓

GUITAR ← VOCAL → KEYBOARD

MONITOR MONITOR

AUDIENCE

MAIN PA L/R

Kun fakkeenya qofa; stage size fi band arrangement irratti hundaa'uun jijjiirama.

17.13. SPEECH STAGE

Nama tokko dubbatu:

SPEAKER

↓
[MIC]

MONITOR

Speech keessatti microphone fi speaker placement sirrii ta'uun feedback fi noise hir'isa.

3. ብብል MIXER IRRATTI AKKAMITTI HOJJETU?

17.14. STAGE IRRAA MIXERITTI

Source tokkoon tokkoon isaa label godhi.

Fakkeenya:

MIC 1 → VOCAL

MIC 2 → VOCAL 2

DI 1 → KEY L

DI 2 → KEY R

MIC 3 → GUITAR AMP

MIC 4 → KICK

MIC 5 → SNARE

17.15. INPUT PATCH

Stage irraa:

MIC
↓
XLR
↓
SNAKE / MULTICORE YOO JIRAATE
↓
MIXER INPUT

Snake ykn multicore fayyadamuun cable hedduu stage irraa mixeritti qindeessuu salphisa.

17.16. MONITOR POSITION

AUX 1 → Vocal Monitor

AUX 2 → Keyboard Monitor

AUX 3 → Band Monitor

AUX output:

MIXER

↓

AUX

↓

MONITOR SPEAKER

17.17. MAIN PA POSITION

MIXER

↓

MAIN L/R

↓

AMPLIFIER / DSP

↓

MAIN SPEAKERS

Speaker audience coverage gaarii akka argatuuf bakka fi olka'iinsa isaa irratti xiyyeeffadhu.

Speaker dhaabbii yeroo fayyadamu, stand ykn mounting system manufacturer's safety requirement irratti hundaa'i.

17.18. MICROPHONE ANGLE

Microphone angle source irratti dhiibbaa qaba.

Vocal keessatti microphone afaanitti kallattii sirriin kaa'amuun:

→ Direct signal fooyyeessa.

Instrument tokko tokko keessatti angle jijjiiruun tone jijjiiruu danda'a.

Kanaaf placement dhaggeeffachuun murteessaadha.

17.19. SOUND SOURCE ISOLATION

Meeshaalee walitti dhihoo jiran irraa bleed uumamuu danda'a.

Fakkeenya:

Guitar Amp → Vocal Mic keessa seenuu danda'a.

Furmaata:

- Amp placement jijjiiri.
- Mic direction sirreessi.
- Level sirreessi.
- Baffles/isolation yeroo barbaachisaa ta'etti fayyadami.

17.20. FINAL STAGE CHECK

Live dura:

- ✓ Microphone hundi bakka isaa jira.
- ✓ Monitor hundi bakka isaa jira.
- ✓ Main speaker nageenyaan dhaabbate.
- ✓ Cable karaa namni irra deemutti hin jirre.
- ✓ Cable labels sirrii.
- ✓ Power connections nageenya qabu.
- ✓ Stands/mounts secure.
- ✓ Instruments sirriitti connect ta'an.

17.1. STAGE SETUP MAAL JECUUDHA?

Stage Setup jechuun meeshaalee sagalee fi performers stage irratti bakka sirrii, nageenya qabu fi hojii isaanii salphisuun qopheessuudha.

Stage setup keessatti:

- Microphone
- Microphone stand
- Monitor
- Keyboard
- Guitar
- Bass
- Drum
- DI Box
- Cables
- Main speaker

fi meeshaaleen biroo bakka isaanii sirriitti qindaa'u.

Setup gaariin:

- Sound qulqulluu
- Feedback xiqqaa
- Cable rakkoo xiqqaa
- Performance mijataa

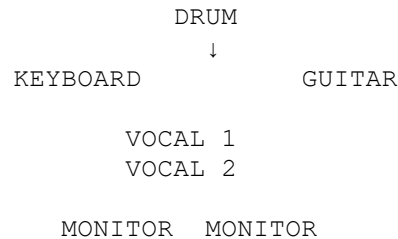
→ Safety gaarii

fida.

17.2. STAGE PLOT

Stage Plot jechuun stage irratti meeshaaleen eessa akka jiran agarsiisuudha.

Fakkeenya:



Stage plot dursee qopheessuun setup saffisa.

17.3. MICROPHONE PLACEMENT

Microphone Placement jechuun microphone source sagalee irraa fageenya fi kallattii sirriin kaa'uudha.

Placement sirrii:

- Signal qulqulluu
- Feedback xiqqaa
- Room noise xiqqaa
- EQ hojii xiqqaa

fiduu danda'a.

17.4. MICROPHONE FI SOURCE

Microphone source irraa:

- Baay'ee fagoo
- Baay'ee dhihoo
- Kallattii dogoggoraa

yoo ta'e sound rakkoo qabaachuu danda'a.

Kanaaf microphone placement source irratti hundaa'a.

17.5. VOCAL MICROPHONE

Vocal microphone yeroo baay'ee:

→ Afaan namaa gara capsule/microphone axisitti

→ Fageenya madaalawaa

→ Monitor speaker irraa kallattii sirrii

irratti kaa'ama.

Singer microphone grille guutuu harkaan cufuu hin qabu.

17.6. KICK DRUM

Kick microphone:

→ Drum keessa ykn port fuulduraa naannoo isaa

→ Source irratti kallattii sirrii

kaa'amuu danda'a.

Bakka microphone kaa'amuun attack fi low-end jijjiiruu danda'a.

17.7. SNARE DRUM

Snare microphone:

→ Snare head naannoo isaa

→ Rim fi head placement irratti hundaa'uun

kaa'amuu danda'a.

Microphone baay'ee centeritti dhiyeessuun sound fi transient jijjiiruu danda'a.

17.8. GUITAR AMP

Guitar amplifier fuuldura microphone kaa'uun:

GUITAR

↓

AMP

↓

MIC

↓

MIXER

Bakka microphone speaker cone irratti kaa'amuun sound jijjiira.

17.9. KEYBOARD

Keyboard yeroo baay'ee:

→ Stand irratti

→ Performer'n sirriitti taphachuu danda'u

→ Cable nama hin gufachii fine

ta'een qophaa'a.

17.10. MONITOR PLACEMENT

Monitor speaker performer fiuldura kaa'ama.

Garuu microphone pickup pattern isaa ilaalamuu qaba.

Kaayyoon:

MONITOR

↓

PERFORMER

fi:

MICROPHONE

↗

UNWANTED FEEDBACK

2. 🎧 FAKKEENYA SAGALEE

17.11. VOCAL MIC

Singer:

“LA LA LA”

Microphone baay'ee dhihoo:

→ Low frequency dabalu danda'a.

Microphone baay'ee fagoo:

→ Room sound dabala.

Fageenya madaalawaan:

→ Vocal natural fi controlled ta'uu danda'a.

17.12. GUITAR AMP MIC

Microphone speaker cone centeritti yoo kaa'ame:

→ Sagalee ifa/attack caalaa argamuu danda'a.

Edge irratti yoo kaa'ame:

→ Sound caalaatti warm ykn mellow ta'uu danda'a.

Kun microphone fi speaker irratti hundaa'uun garaagarummaa qaba.

17.13. DRUM MIC

Kick:

“BOOM”

Snare:

“CRACK”

Cymbal:

“CHH”

Microphone tokkoon tokkoon isaa source isaa irratti xiyyeeffachuun bleed hir'isuuf gargaara.

17.14. MONITOR FI FEEDBACK

MIC

↓

MIXER

↓

MONITOR

↓

MIC

Yoo monitor irraa sagaleen gara microphone deebi'e:

“IIIIII!”

Feedback uumamuu danda'a.

Kanaaf monitor placement baay'ee barbaachisaa dha.

17.15. CABLE ROUTING

Cable stage irratti:

SOURCE

↓

CABLE



STAGE BOX / SNAKE



MIXER

Karaa namni irra deemu irraa fageessuun safety fooyyeessa.

3. ብቅ MIXER IRRATTI AKKAMITTI HOJJETU?

17.16. CHANNEL MAPPING

Stage setup booda channel assignment qopheessi.

Fakkeenya:

CH1 → Lead Vocal

CH2 → Vocal 2

CH3 → Keyboard L

CH4 → Keyboard R

CH5 → Guitar

CH6 → Bass

CH7 → Kick

CH8 → Snare

Kun mixer irratti channel barbaaduu salphisa.

17.17. INPUT GAIN

Microphone placement erga sirrii ta'ee:

→ Performer'n source isaa haa taphatu/dubbatu.

→ PFL/Solo fayyadami.

→ Gain suuta qindeessi.

→ Clipping irraa fagaadhu.

17.18. MONITOR AUX

Stage setup keessatti monitor tokko tokkoof AUX adda addaa fayyadamuun ni danda'ama.

Fakkeenya:

AUX 1 → Vocal Monitor

AUX 2 → Keyboard Monitor

AUX 3 → Band Monitor

Performer tokko maal akka barbaadu irratti hundaa’uun send qindeessi.

17.19. STAGE VOLUME

Stage volume baay’ee ol yoo ta’e:

- Main mix rakkoo qabaachuu danda’a.
- Microphone bleed dabala.
- Feedback risk dabala.
- Sound Engineer’n control dhabuu danda’a.

Kanaaf stage volume controlled ta’uu qaba.

17.20. SOUND CHECK

Stage setup xumuramee booda:

1. Microphone test.
2. Instrument test.
3. Monitor test.
4. Main PA test.
5. Feedback check.
6. Cable check.
7. Final mix check.
8.   DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — MICROPHONE MONITOR FUULDURA KAAUU

Feedback fiduu danda’a.

Furmaata:

Pickup pattern fi monitor position ilaali.

DOGOGGORA 2 — MICROPHONE HEAD HARKAAN CUGUU

Polar pattern jijjiiruu fi feedback dabalu danda’a.

Furmaata:

Body microphone irraa qabadhu.

DOGOGGORA 3 — CABLE KARA NAMNI IRRA DEEMU KEESSA DIRIIRSUU

Nama gufachiisuu ykn cable miidhuu danda'a.

Furmaata:

Cable route sirreessi; yeroo barbaachisaa ta'etti cable ramp ykn cover fayyadami.

DOGOGGORA 4 — MONITOR HUNDAA GARA TOKKOTTI QOFA QABUU

Performer hundi bakka adda addaatti jiraachuu danda'a.

Furmaata:

Monitor coverage fi performer position ilaali.

DOGOGGORA 5 — STAGE VOLUME BAAY'EE OLKAASUU

Kun main mix fi feedback irratti rakkoo fida.

Furmaata:

Stage volume gadi-aanaa fi controlled eegi.

DOGOGGORA 6 — MICROPHONE HUNDAA WAL QIXA FAGEESSUU

Source hundi sensitivity fi volume wal fakkaataa hin qabu.

Furmaata:

Source tokkoon tokkoon isaa irratti placement sirreessi.

DOGOGGORA 7 — STAGE SETUP BOODA LINE CHECK DHIISUU

Setup yeroo socho'u cable ykn connector tokko rakkoo qabaachuu danda'a.

Furmaata:

Setup booda line check deebisanii godhi.

DOGOGGORA 8 — SAFETY DAGATUU

Stand kufuu, cable gufachiisuu ykn equipment sirriitti hin kaa'amne balaa fiduu danda'a.

Furmaata:

Stage setup keessatti **sagalee qulqulluu qofa osoo hin taane safety** dursa kenni.

YAADA IJOO

STAGE SETUP

→ MEESHAALÉE BAKKA SIRRIITTI QOPHEESSUU

MIC PLACEMENT

→ SOURCE IRRATTI KALLATTII FI FAGEENYA SIRRII

MONITOR PLACEMENT

→ PERFORMER + FEEDBACK CONTROL

CABLE ROUTING

→ SIGNAL + SAFETY

STAGE VOLUME

→ MIX CONTROL

Sound Engineer professional tokko mixer dura **stage irratti** rakkoo hedduu furu danda'a. Microphone, monitor, cable fi performer bakka sirriitti yoo qindaa'an, mixer irratti hojii baay'een salphata.

BOQONNAA 18 — LIVE MIXING: YEROO PERFORMANCE SAGALEE TO'ACHUU

1. IBSA BAL'AA

18.1. LIVE MIXING MAAL JECUUDHA?

Live Mixing jechuun performance yeroo adeemaa jiru keessatti sagalee microphone, instrument fi effects hunda walitti madaalchuun audience'ni sagalee qulqulluu fi madaalawaa akka dhaga'u gochuudha.

Sound Engineer yeroo live:

- Volume
- Balance
- EQ
- Monitor
- Effects
- Feedback
- Mute
- Dynamics

to'ata.

18.2. MIX BALANCE

Mix balance jechuun source tokko kan biraa akka hin haguugne qindeessuudha.

Fakkeenya:

VOCAL → ifa

KEYBOARD → vocal hin haguugu

GITAR → keyboard hin haguugu

BASS → low-end guuta

DRUM → rhythm ifa

Kaayyoon isaa source hunda olkaasuu miti; **source barbaachisaa tokkoon tokkoon isaa bakka isaa akka qabaatu gochuudha.**

18.3. FOCAAL POINT

Performance keessatti yeroo hunda wanti audience'n xiyyeeffatu jira.

Fakkeenya:

Sirba keessatti:

→ Lead Vocal

Instrumental keessatti:

→ Solo instrument

Speech keessatti:

→ Speaker

Sound Engineer'n yeroo barbaachisaa ta'etti focus kana ifa godha.

18.4. VOLUME CONTROL

Live mixing keessatti fader'n yeroo hunda bakka tokko hin turu.

Fakkeenya:

Verse

→ Vocal level normal

Chorus

→ Vocal ykn instruments level jijjiiramuu danda'a

Solo

→ Solo instrument olka'uu danda'a

Kun **mix automation harkaan** fakkaata.

18.5. MUTE

Mute jechuun channel tokko main mix irraa yeroo muraasaaf dhoksuudha.

Fakkeenya:

Microphone hin fayyadamne:

→ MUTE

Kun:

- Stage noise
- Feedback risk
- Unwanted bleed

hir'isuuf gargaara.

18.6. GROUP / BUS

Mixer tokko tokko group/bus qaba.

Fakkeenya:

VOCAL 1

VOCAL 2

VOCAL 3

↓

VOCAL BUS

↓

MAIN MIX

Kun vocal hundaa yeroo tokkootti to'achuuf gargaara.

18.7. DYNAMICS

Dynamics jechuun sagalee laafaa fi jabaa gidduu garaagarummaa level ti.

Sound Engineer:

→ Level baay'ee ol akka hin baane

→ Sagaleen lafaan akka hin badne

to'achuu danda'a.

Compressor ykn limiter yoo jiraate sirriitti fayyadamuun barbaachisaa dha.

18.8. REVERB FI DELAY

Reverb:

→ Sagalee room ykn space fakkeessa.

Delay:

→ Sagalee irra deddeebii yeroo murtaa'e booda dhaga'amu uuma.

Vocal irratti yeroo baay'ee effect kana fayyadamuun ni danda'ama.

Garuu effect baay'inaan:

→ Vocal clarity hir'isa.

2. 🎧 FAKKEENYA SAGALEE

18.9. VERSE → CHORUS

VERSE:

Vocal

→ Normal

Band

→ Moderate

CHORUS:

Vocal

→ Controlled boost

Band

→ Full

Kaayyoon chorus gara verse irraa adda akka dhaga'amu gochuudha.

18.10. SOLO

Guitarist:

“SOLO!”

Sound Engineer:

Guitar fader

↑

Vocal fi instruments biroo balance isaanii eegu.

Guitar solo audience'fi ifa ta'a.

18.11. VOCAL OVER MUSIC

Yoo keyboard ykn guitar vocal haguugu:

Vocal ↑

ykn:

Instrument ↓

EQ irratti frequency masking rakkoo ta'e sirreessuunis ni danda'ama.

18.12. BASS FI KICK

Kick:

“BOOM”

Bass:

“BUUUM”

Lamaan isaanii low-end keessatti walitti bu'uu danda'u.

Sound Engineer:

→ EQ

→ Level

→ Arrangement

fayyadamuun separation fooyyeessa.

18.13. SPEECH MIX

Speech keessatti:

VOICE

→ Main focus

Music

→ Gadi-aanaa

Reverb

→ Xiqqoo ykn hin barbaachisu

Kaayyoon:

Jecha tokkoon tokkoon isaa ifatti dhaga'amu.

18.14. WORSHIP MIX

Worship keessatti:

Lead vocal

→ Focus

Backing vocal

→ Support

Keyboard

→ Support

Guitar

→ Support

Bass

→ Foundation

Drums

→ Rhythm

Mix'n namni sirba fi ergaa isaa hordofuu danda'u ta'uu qaba.

3. MIXER IRRATTI AKKAMITTI HOJJETU?

18.15. STARTING FADER POSITION

Soundcheck booda channel fader bakka hojii madaalawaa irraa jalqabi.

Fakkeenya:

Vocal → Main focus

Keyboard → Support

Guitar → Support

Bass → Foundation

Drums → Rhythm

Lakkoofsa tokko qofa akka seeraatti hin fudhatin; mixer, room fi source irratti hundaa'i.

18.16. CHANNEL SOLO

Channel tokko qofa dhaggeeffachuuf:

→ SOLO/PFL

fayyadami.

Garuu mix dhumaa yeroo hunda **full mix** keessatti murteessi.

18.17. FADER MOVEMENT

Live keessatti:

Vocal gadi bu'e:

→ Fader xiqqoo olkaasi.

Vocal baay'ee olka'e:

→ Fader xiqqoo gadi buusi.

Solo:

→ Solo instrument boost godhi.

Fader saffisaan ol-gadi gochuu irraa of eegi.

18.18. PAN

Stereo mixer keessatti:

PAN L ← CENTER → R

Keyboard stereo fakkeenyaaf:

Keyboard L → Left

Keyboard R → Right

Pan baay'ee gara cinaacha tokkootti gochuun balance room irratti rakkoo fiduu danda'a.

18.19. FX SEND

Vocal:

CHANNEL

↓

AUX/FX SEND

↓

REVERB

↓

RETURN

↓

MAIN MIX

FX level suuta dabali.

Kaayyoon effect dhaga'amu osoo hin taane **vocal deeggaruu** ta'uu danda'a.

18.20. FEEDBACK YEROO LIVE

Yoo:

“IIIIII!”

ykn “WOOOOO!”

dhaga'ame:

1. Level rakkoo qabu saffisaan gadi buusi.
2. Microphone ykn monitor adda baasi.
3. Frequency rakkoo barbaadi.
4. EQ/graphic EQ irratti xiqqoo cut godhi.
5. Mic fi speaker placement sakatta'i.

Feedback yeroo live keessatti saffisaan to'achuun hojii Sound Engineer keessaa isa guddaadha.

4. △□ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — CHANNEL HUNDAA OLKAASUU

Channel hundi ol yoo ta'e mix:

→ Loud

→ Muddy

→ Unclear

ta'uu danda'a.

Furmaata:

Balance godhi.

DOGOGGORA 2 — VOCAL HAGUU

Music baay'ee ol yoo ta'e vocal dhaga'amuun rakkisa.

Furmaata:

Vocal focus godhi; music level/EQ sirreessi.

DOGOGGORA 3 — EFFECT BAAY'EE

Reverb baay'inaan vocal akka fagoo fi ifa hin taane gochuu danda'a.

Furmaata:

Dry/Wet balance dhaggeeffadhu.

DOGOGGORA 4 — FADER TASAA OLKAASUU

Audience'fi nama stage irratti shock uumuu danda'a.

Furmaata:

Fader suuta sochoosi.

DOGOGGORA 5 — FEEDBACK YEROO JIRAATU EQ QOFAAN TURUU

Feedback yeroo live saffisaan to'atamuu qaba.

Furmaata:

Dura level/mic/monitor sirreessi; booda EQ.

DOGOGGORA 6 — SOLO KEESSATTI MIX MURTEESSUU

Solo keessatti source tokko gaarii fakkaachuu danda'a.

Garuu full mix keessatti rakkoo qabaachuu danda'a.

Furmaata:

Solo → qorannoo.

Full Mix → murtii.

DOGOGGORA 7 — AUDIENCE DHAGGEEFFACHUU DHIISUU

Booth ykn mixer biraa irraa sound gaarii fakkaachuu danda'a; audience bakka adda addaatti garuu garaagarummaa dhaga'a.

Furmaata:

Yeroo danda'ametti room keessa bakka adda addaa dhaggeeffadhu.

DOGOGGORA 8 — SOUND ENGINEER PERFORMANCE IRRATTI XIYYEEFFATUU DHIISUU

Sound Engineer yeroo live:

→ Singer

→ Band

→ Speaker

→ Audience

hunda hordofuu qaba.

YAADA IJOO

LIVE MIXING

→ SAGALEE YEROO PERFORMANCE TO'ACHUU

BALANCE

→ SOURCE HUNDAA WAL SIMSIISUU

FADER

→ LEVEL TO'ACHUU

MUTE

→ CHANNEL DHOKSUU

PAN

→ STEREO POSITION

FX

→ SPACE / EFFECT

FEEDBACK

→ RAKKOO SAFFISAAN TO'ACHUU

Sound Engineer professional tokko **mixer qofa hin ilaalu; gurraan dhaggeeffata, meter ilaala, performance hordofa, audience ilaala, yeroo sirriitti murtii fudhata**

BOQONNAA 19 — TROUBLESHOOTING: RAKKOO SAGALEE ADDAA BAASUU FI SIRREESSUU

1. IBSA BAL'AA

19.1. TROUBLESHOOTING MAAL Jechuudha?

Troubleshooting jechuun yeroo sound system keessatti rakkoon uumamu, sababaa rakkoo sanaa sirnaan barbaaduu fi tarkaanfii sirrii fudhachuudha.

Sound Engineer tokko rakkoo yoo arge:

“Maaliif sagaleen hin jiru?”

qofa jechuu hin qabu.

Gaaffii kana gaafachuu qaba:

“Signal eessa irraa dhufaa jira?”

“Eessa irratti bade?”

“Rakkoon cable moo mixer moo amplifier moo speaker?”

Kana jechuun rakkoo tokko **tartiibaan adda baasuu** dha.

19.2. SIGNAL FLOW HUBACHUU

Troubleshooting keessatti signal flow beekuun murteessaa dha.

SOURCE

↓

MIC / INSTRUMENT

↓

CABLE

↓

MIXER INPUT

↓

GAIN

↓

CHANNEL

↓

BUS

↓

MAIN OUTPUT

↓

AMPLIFIER / POWERED SPEAKER

↓

SPEAKER

↓
AUDIENCE

Sagaleen yoo hin jirre, tartiiba kana hordofuun bakka signal itti bade barbaadi.

19.3. “NO SOUND” RAKKOO

Yoo microphone irraa sagaleen hin dhufne:

1. Microphone ilaali.
2. Cable ilaali.
3. Mixer input ilaali.
4. Channel mute ilaali.
5. Gain ilaali.
6. PFL/Solo ilaali.
7. Fader ilaali.
8. Main routing ilaali.
9. Main output ilaali.
10. Amplifier/speaker ilaali.

Tokko tokkoon qorachuun bakka rakkoo adda baasa.

19.4. HUM MAAL JECUUDHA?

Hum jechuun sagalee:

“mmmmmmmm”

fakkaatuudha.

Sababoota isaa keessaa:

- Grounding rakkoo
- Cable rakkoo
- Power interference
- Unbalanced connection dheeraa
- Equipment walitti dhufeenya rakkoo qabu

ta’uu danda’a.

Hum yeroo baay’ee power system fi audio wiring sirriitti qorachuu barbaada.

19.5. BUZZ / NOISE

Buzz ykn unwanted noise yoo jiraate:

→ Cable

→ Connector

→ Power

→ Gain

→ External equipment

sakatta'i.

Rakkoo sana adda baasuuf source tokko yeroo tokkootti isolate godhuun gargaara.

19.6. DISTORTION

Sagaleen:

“CRACK”

“FUZZ”

“BREAKING”

fakkaatu yoo ta'e distortion jiraachuu danda'a.

Sababoota:

- Input gain baay'ee ol
- Mixer output clipping
- Amplifier input overload
- Speaker overload

ta'uu danda'u.

Bakka distortion itti uumamu adda baasi.

19.7. FEEDBACK

Feedback:

“IIIIII!”

“WOOOO!”

fakkaachuu danda'a.

Signal loop:

MIC

↓

MIXER

↓

MONITOR / SPEAKER

↓
MIC
↓
MIXER

Yoo loop kun frequency tokko irratti of cimsu:

→ Feedback.

19.8. INTERMITTENT SIGNAL

Sagaleen yeroo tokko jira, yeroo biraa bada yoo ta'e:

→ Cable

→ Connector

→ Patch

→ Input jack

→ Microphone

sakatta'i.

Kun yeroo baay'ee connection rakkoo qabu irraa dhufuu danda'a.

19.9. LEFT / RIGHT RAKKOO

Stereo system keessatti:

Left jira.

Right hin jiru.

Yoo ta'e:

1. Cable R ilaali.
2. Mixer R output ilaali.
3. Pan ilaali.
4. Amplifier CH2 ilaali.
5. Right speaker ilaali.

Tartiibaan qoradhu.

19.10. LOW FREQUENCY HIN JIRRE

Bass ykn kick gadi fagoo hin dhaga'amne yoo ta'e:

- HPF ilaali.
- EQ ilaali.
- Speaker/subwoofer connection ilaali.
- Crossover ilaali.
- Polarity/phase relationship ilaali.
- Bass channel level ilaali.

Frequency low bade jechuun speaker qofa rakkoo qaba jechuu miti.

2. 🎧 FAKKEENYA SAGALEE

19.11. MICROPHONE SAGALEE HIN QABNE

MIC

↓

CABLE

↓

MIXER

PFL irratti:

→ Signal hin jiru.

Tartiiba:

MIC biraa yaali.

Yoo mic haaraan hojjetee:

→ Mic duraa rakkoo qabaachuu danda’a.

Yoo lamaan isaanii hin hojjenne:

→ Cable/input/power/patch sakatta’i.

19.12. CABLE RAKKOO

Mic tokko:

“Check... check...”

Yeroo tokko dhaga’ama.

Yeroo biraa bada.

Cable sochoosuun:

“crack crack”

fakkaata.

Kun cable ykn connector rakkoo qabaachuu danda'a.

Cable sana hojii professional keessatti itti fufsiisuu irra bakka buusuun filannoo nageenya qabuudha.

19.13. AMPLIFIER OUTPUT RAKKOO

Mixer output jira.

Amplifier input meter jira.

Garuu speaker irraa sagalee hin jiru.

Tartiiba:

AMPLIFIER OUTPUT

↓

SPEAKER CABLE

↓

SPEAKER INPUT

↓

SPEAKER

Qoradhu.

19.14. ONE SPEAKER ONLY

Left speaker jira.

Right speaker hin jiru.

Qorannoo:

Mixer R

→ cable R

→ amplifier CH2

→ speaker cable

→ right speaker

Bakka signal itti bade adda baasi.

19.15. FEEDBACK YEROO SIRBAA

Singer microphone qabatee:

“AAAAAA!”

Monitor:

“IIIIII!”

Yeroo kana:

1. Monitor level gadi buusi.
2. Vocal channel ilaali.
3. Mic position sirreessi.
4. Monitor placement ilaali.
5. Frequency rakkoo adda baasi.
6. EQ/graphic EQ irratti xiqqoo cut godhi.

19.16. HUM YEROO KEYBOARD CONNECT GODHU

Keyboard connect godhamee:

“MMMMMMMM”

dhaga’ama.

Qoradhu:

Keyboard output

↓

Cable

↓

DI box / input

↓

Mixer

Cable biraa ykn DI box sirrii yaali.

Power arrangement fi grounding hojii isaa ogeessa gahumsa qabuun qorachiisi.

19.17. DISTORTED VOCAL

Singer yeroo sagalee olkaasu:

“AAAA!” → distorted

Yeroo sagalee gadi qabu:

→ clean

Kun input gain ykn signal stage tokko overload ta’uu danda’a.

Gain input sakatta’i.

19.18. METER GAARII, SAGALEEN GARUU HIN JIRU

Meter signal agarsiisa.

Garuu speaker irraa sound hin dhaga’amu.

Kun:

→ Routing

→ Output

→ Amplifier

→ Speaker

keessa rakkoo ta'uu danda'a.

Meter tokko qofa ilaaluun troubleshooting xumuruun sirrii miti.

3. ብቅዝ MIXER IRRATTI AKKAMITTI HOJJETU?

19.19. TROUBLESHOOTING TARTIIBA

Yeroo rakkoon uumamu:

STEP 1

→ CALM TA'I.

STEP 2

→ RAKKOO MAAL AKKA TA'E ADDABAASI.

STEP 3

→ SIGNAL FLOW YAADI.

STEP 4

→ SOURCE IRRAA JALQABI.

STEP 5

→ INPUT QORI.

STEP 6

→ MIXER QORI.

STEP 7

→ OUTPUT QORI.

STEP 8

→ AMPLIFIER/SPEAKER QORI.

STEP 9

→ RAKKOO SIRREESSI.

STEP 10

→ SYSTEM DEEBI'EE TEST GODHI.

19.20. MUTE CHECK

Channel tokko sagalee hin qabne yoo ta'e:

→ MUTE button ilaali.

Main mute ykn group mute yoo jiraate illee ilaali.

19.21. PHANTOM POWER

Condenser microphone tokko tokko phantom power barbaadu.

Mixer irratti:

+48V

jedhamee mul'achuu danda'a.

Garuu phantom power equipment hundarraa wal qixa hin eegalin.

Fakkeenyaaf ribbon microphone passively designed tokko irratti phantom power connection dogoggoraa ykn cable rakkoo qabu miidhaa fiduu danda'a.

Kanaaf microphone manufacturer's instruction hordofi.

19.22. PFL QORANNOO

Channel PFL/Solo yoo qabate:

Channel

↓

PFL

↓

Meter

Signal input jiraachuu isaa ilaali.

PFL irratti signal jira:

→ Rakkoon channel booda ta'uu danda'a.

PFL irratti signal hin jiru:

→ Rakkoon source/input/gain/cable ta'uu danda'a.

19.23. MAIN OUTPUT QORANNOO

Channel meter jira.

Main meter hin jiru.

Kana jechuun:

→ Bus routing

→ Channel assignment

→ Main mix

sakatta'i.

Main meter jira.

Amplifier input hin jiru.

Kana jechuun:

→ Output cable

→ Amplifier input

sakatta'i.

19.24. ISOLATION METHOD

Rakkoo adda baasuuf equipment tokko tokko yeroo muraasaaf system keessaa baasuu dandeessa.

Fakkeenya:

Source A → test

Source B → test

Cable A → test

Cable B → test

Kun rakkoo eessa akka jiru adda baasuuf gargaara.

19.25. BACKUP PLAN

Professional Sound Engineer yeroo hunda backup qabaachuu qaba.

Fakkeenya:

- Spare XLR cable
- Spare microphone
- Spare battery
- Spare adapter

- Backup playback source
- Spare fuse yoo manufacturer-approved replacement ta'e

Equipment bakka buusuun yeroo rakkoo live keessatti baay'ee gargaara.

4. $\Delta \square$ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — RAKKOO YEROO TOKKOON EQUIPMENT HUNDAA JIJJIIRUU

Kun rakkoo dhugaa eessa akka jiru adda baasuu rakkisa.

Furmaata:

Tartiibaan qoradhu.

DOGOGGORA 2 — CABLE QOFA SHAKKUU

Rakkoon cable qofa miti.

Source, mixer, routing, amplifier fi speaker hundi sababaa ta'uu danda'u.

DOGOGGORA 3 — FEEDBACK YEROO JIRAATU LEVEL OLKAASUU

Kun rakkoo caalaatti hammeessa.

Furmaata:

Level gadi buusi.

DOGOGGORA 4 — HUM YOO JIRAATE GROUND CONNECTION OFII KEETIIN JIJJIIRUU

Electrical grounding hojii balaa qabu ta'uu danda'a.

Furmaata:

Electrical system nama gahumsa qabuun haa qoramu.

DOGOGGORA 5 — CLIPPING METER IRRATTI ARGAMUU DAGATUU

Meter red/clip yoo agarsiise:

→ Signal stage tokko overload ta'uu danda'a.

Bakka clipping itti uumamu barbaadi.

DOGOGGORA 6 — RAKKOO SIRREESSUU JETTEE PERFORMANCE GUUTUU DHAABUU

Rakkoo xiqqaa yoo ta'e:

→ Mute

→ Backup

→ Channel biraa

fayyadamuun performance itti fufsiisuun ni danda'ama.

Garuu safety ykn equipment damage risk yoo jiraate, system nageenyaan dhaabi.

DOGOGGORA 7 — SAFETY DHIISUU

Cable, electricity, speaker rigging ykn amplifier rakkoo qabu irratti yaalii balaa qabu hin godhin.

Hojii mains electricity fi rigging nama ogummaa qabuuf dhiisi.

YAADA IJOO

TROUBLESHOOTING

→ RAKKOO TARTIIBAAN ADDA BAASUU

NO SOUND

→ SIGNAL FLOW HORDOFI

HUM

→ CABLE / POWER / GROUNDING SAKKATTA'I

DISTORTION

→ GAIN / OUTPUT / AMPLIFIER SAKKATTA'I

FEEDBACK

→ MIC + SPEAKER + LEVEL + FREQUENCY

ONE SIDE OUT

→ L/R SIGNAL CHAIN QORI

INTERMITTENT

→ CABLE + CONNECTOR + PATCH QORI

PFL

→ INPUT SIGNAL QORUUF

BACKUP

→ LIVE PERFORMANCE EEGUUF

Sound Engineer professional tokko rakkoo yeroo uumamu **rifachuu hin qabu**. Signal flow yaadee, source irraa jalqabee, bakka signal itti bade adda baasee, rakkoo sana sirnaan fura.

BOQONNAA 20 — PROFESSIONAL LIVE SOUND WORKFLOW

1. IBSA BAL'AA

20.1. PROFESSIONAL SOUND ENGINEER MAAL HOJJECHA?

Sound Engineer professional jechuun mixer qofa hojjechuu miti.

Inni:

- Performance dura system qopheessa.
- Stage setup ilaala.
- Microphone fi cable qindeessa.
- Line check godha.
- Soundcheck gaggeessa.
- Gain staging sirreessa.
- EQ qindeessa.
- Monitor mix hojjeta.
- Main PA to'ata.
- Yeroo performance mix godha.
- Feedback fi rakkoo sagalee saffisaan fura.
- Performance booda system sirnaan cufa.

Kanaaf hojii isaa:

PREPARATION

↓

SETUP

↓

LINE CHECK

↓

SOUNDCHECK

↓

LIVE MIXING

↓

TROUBLESHOOTING

↓

SHUTDOWN

20.2. PERFORMANCE DURA

Performance dura Sound Engineer'n:

1. Venue ilaala.
2. Stage size hubata.
3. Performers meeqa akka jiran beeka.
4. Input list qopheessa.
5. Stage plot ilaala.
6. Equipment inventory godha.
7. Cable fi connector qopheessa.

8. Mixer setup godha.
9. Main PA qopheessa.
10. Monitor qopheessa.

20.3. INPUT LIST

Input list fakkeenya:

CH1 — Lead Vocal

CH2 — Backing Vocal

CH3 — Keyboard L

CH4 — Keyboard R

CH5 — Guitar

CH6 — Bass

CH7 — Kick

CH8 — Snare

CH9 — Overhead L

CH10 — Overhead R

CH11 — Playback L

CH12 — Playback R

Input list'n Sound Engineer channel hunda dursee beekuuf gargaara.

20.4. STAGE SETUP

Stage irratti:

- Microphone
- Stands
- DI boxes
- Monitors
- Instruments
- Cables

qindaa'u.

Cable karaa namni irra deemu irraa eegi.

Speaker stands ykn rigging sirnaan secure ta'uu qabu.

20.5. SYSTEM CHECK

System guutuu:

SOURCE

↓

MIXER

↓

MAIN OUTPUT

↓

AMPLIFIER / DSP

↓

SPEAKER

tokko tokkoon qoradhu.

Sagaleen bakka hundatti darbu isaa mirkaneessi.

20.6. LINE CHECK

Channel tokkoon tokkoon isaa:

SOURCE

↓

INPUT

↓

METER

qoradhu.

Line check booda rakkoon tokko yoo jiraate performance dura furi.

20.7. SOUNDCHECK

Performer tokkoon tokkoon isaa waliin:

→ Gain

→ EQ

→ Fader

→ Monitor

→ Main mix

qindeessi.

Soundcheck keessatti sagalee performance dhugaa fakkaatu fayyadamuun gaarii dha.

20.8. MONITOR MIX

Performer tokko tokko wanta adda addaa barbaadu.

Fakkeenya:

Singer:

→ Vocal isaa

→ Keyboard xiqqoo

→ Guitar xiqqoo

Keyboardist:

→ Vocal

→ Keyboard

→ Band

Monitor mix tokko hundumaaf wal fakkaachuu hin qabu.

20.9. MAIN MIX

Main mix audience'fi qophaa'a.

Kaayyoo:

VOCAL → CLEAR

MUSIC → BALANCED

BASS → CONTROLLED

DRUM → CLEAR

HIGH → NOT HARSH

OVERALL → NATURAL

2. 🎧 FAKKEENYA SAGALEE

20.10. LIVE WORSHIP

Fakkeenya setup:

VOCAL



MIXER CH1

KEYBOARD



CH2/3

GUITAR



CH4

BASS



CH5

DRUM



CH6–10

MIXER



MAIN L/R



PA SYSTEM

20.11. SPEECH EVENT

Speaker:

MIC



MIXER



MAIN OUTPUT



PA SPEAKER

Speech keessatti:

→ Vocal clarity

→ Feedback control

→ Appropriate volume

irratti xiyyeeffadhu.

20.12. BAND PERFORMANCE

Performance jalqabee:

VERSE

↓

BALANCED MIX

CHORUS

↓

FULLER MIX

SOLO

↓

SOLO INSTRUMENT FOCUS

ENDING

↓

CONTROLLED FADE / FINAL LEVEL

20.13. FEEDBACK YEROO PERFORMANCE

Feedback yoo dhaga'ame:

“IIIII!”

Tarkaanfii jalqabaa:

→ Level rakkoo qabu gadi buusi.

Booda:

→ Mic placement

→ Monitor level

→ EQ frequency

sakatta'i.

20.14. RAKKOO MICROPHONE

Mic tokko yeroo performance keessa badde:

1. Mute godhi.
2. Backup mic qopheessi.
3. Cable/connection qoradhu.
4. Channel biraa yoo barbaachise fayyadami.
5. Performance akka hin jeeqamne saffisaan sirreessi.

20.15. POWER RAKKOO

Power rakkoon yoo uumame:

→ Safety dursa.

Electrical system irratti yaalii balaa qabu hin godhin.

Ogeessa electrical fi venue safety procedure hordofi.

20.16. AUDIENCE COVERAGE

Main PA qofa mixer biraa irraa dhaggeeffachuun gahaa miti.

Audience bakka gara garaa irraa:

→ Volume

→ Clarity

→ Bass

→ High

→ Stereo/coverage

dhaggeeffadhu.

Venue keessa bakka tokko gaarii ta'ee bakka biraa rakkoo qabaachuu danda'a.

3. □ MIXER IRRATTI AKKAMITTI HOJJETU?

20.17. MIXER STARTUP

Setup xumuramee booda:

1. Channel labels mirkaneessi.
2. Gain sirriitti qopheessi.
3. EQ basic qopheessi.
4. Monitor AUX qopheessi.
5. FX qopheessi.
6. Main output qopheessi.
7. Amplifier/powered speaker level manufacturer recommendation irratti qindeessi.
8. System test godhi.

20.18. LIVE PERFORMANCE

Yeroo performance:

→ Gurraan dhaggeeffadhu.

→ Meter ilaali.

→ Fader ilaali.

→ Performers ilaali.

→ Stage ilaali.

→ Audience reaction hubadhu.

Sound Engineer tokko yeroo live **murtii saffisaa fi xiqqoo** fudhata; jijjiirama guddaa tasaa irraa of eega.

20.19. SCENE / SETTING NOTE

Analog mixer keessatti settings diijitaalaa akka “scene recall” hin qabaatu.

Kanaaf:

- Channel list
- Tape/label
- Patch sheet
- Notebook
- Photo/reference

fayyadamuun settings yaadachuuf gargaara.

Mixer analog keessatti documentation baay’ee barbaachisaa dha.

20.20. PERFORMANCE BOODA

Performance erga xumuramee booda:

1. Main level to’achuun sirna nageenya qabuun system cufi.
2. Amplifier/powered speakers dursa cufi.
3. Mixer fi processors itti aansee cufi.
4. Microphones walitti qabi.
5. Cables sirnaan coil godhi.
6. Equipment inventory godhi.
7. Cable ykn equipment miidhame galmeessi.
8. Equipment bakka isaa kaa’i.

20.21. PROFESSIONAL CHECKLIST

PERFORMANCE DURA:

☐ Input list

☐ Stage plot

☐ Microphones

- ☐ Cables
- ☐ DI boxes
- ☐ Monitors
- ☐ Main speakers
- ☐ Mixer
- ☐ Amplifier
- ☐ Power/safety check
- ☐ Line check
- ☐ Soundcheck

PERFORMANCE YEROO:

- ☐ Gain
- ☐ Fader
- ☐ Main mix
- ☐ Monitor
- ☐ Feedback
- ☐ Audience coverage
- ☐ Performers

PERFORMANCE BOODA:

- ☐ System shutdown
- ☐ Cable collection
- ☐ Microphone collection
- ☐ Equipment inventory
- ☐ Damage check

□ Storage

4. △□ DOGOGGORA IRRAA OF EEGUU

DOGOGGORA 1 — PREPARATION DHIISUU

Performance dura qophii gahaa hin goone.

→ Rakkoo live dabala.

Furmaata:

Dursee qophaa'i.

DOGOGGORA 2 — SOUNDCHECK XIQQOO GODHUU

Soundcheck gahaa hin taane:

→ Feedback

→ Bad monitor

→ Poor mix

fiduu danda'a.

DOGOGGORA 3 — PERFORMANCE JALQABEE MIX GUUTUU JIJJIIRUU

Mix yeroo live keessatti jijjiirama guddaa tasaa gochuun rakkoo fida.

Furmaata:

Xiqqoo xiqqoon sirreessi.

DOGOGGORA 4 — AUDIENCE DHAGGEEFFACHUU DHIISUU

Mixer bakka tokko irraa gaarii fakkaachuu danda'a.

Audience coverage garuu bakka gara garaatti adda ta'a.

Furmaata:

Room keessa bakka gara garaa qoradhu.

DOGOGGORA 5 — BACKUP QABAACHUU DHIISUU

Cable tokko yoo bade performance dhaabachuu danda'a.

Furmaata:

Spare cable fi backup equipment barbaachisaa qabaadhu.

DOGOGGORA 6 — EQUIPMENT HUMNAAN HOJJETUU

Knob, connector ykn cable humnaan qabachuun equipment miidhuu danda’a.

Furmaata:

Equipment sirnaan qabsiisi.

DOGOGGORA 7 — SAFETY DAGATUU

Electricity, speaker stands, rigging fi cable safety dhimma guddaa dha.

Furmaata:

Safety yeroo hunda dursa.

DOGOGGORA 8 — BARACHUU DHAABUU

Sound Engineering teknooloojii fi muuziqaa waliin yeroo hunda guddachaa deema.

Professional Sound Engineer:

→ Dhaggeeffata.

→ Shaakala.

→ Qorata.

→ Dogoggora irraa barata.

→ Equipment haaraa hubata.

YAADA IJOO

PROFESSIONAL LIVE SOUND WORKFLOW:

PREPARE

↓

SETUP

↓

PATCH

↓

LINE CHECK

↓

GAIN STAGING

↓

EQ

↓

MONITOR MIX



SOUNDCHECK



MAIN MIX



LIVE PERFORMANCE



TROUBLESHOOTING



SAFE SHUTDOWN



PACK & CHECK

XUMURA KITAABAA

Sound Engineer professional ta'uuf mixer qofa beekuun gahaa miti.

Ati:

✓ Audio fundamentals

✓ Signal flow

✓ Gain staging

✓ EQ

✓ Microphone

✓ Cable & connector

✓ Stage setup

✓ Monitor

✓ Amplifier

✓ Speaker

✓ Soundcheck

✓ Live mixing

✓ Troubleshooting

✓ Safety

✓ Professional workflow

hunda walitti qabdee shaakalu qabda.

Sound Engineer gaarii tokko sagalee qofa hin olkaasu; sagalee qulqulluu, madaalawaa, amanamaa fi audience'fi performer hundaaf mijataa uuma.